

Fractional Calculus and Integration of Deterministic Functions with Respect to FBM

There are fundamental connections between self-similarity and fractional calculus, which is an area of real analysis. These connections are explored here in the context of fractional Brownian motion (FBM).

The basics of fractional calculus can be found in Section 6.1 where we introduce the following fractional integrals and derivatives:

- Riemann–Liouville fractional integral on an interval (Definition 6.1.1)
- Riemann–Liouville fractional derivative on an interval (Definition 6.1.4)
- Fractional integral on the real line (Definition 6.1.13)
- Liouville fractional derivative on the real line (Definition 6.1.14)
- Marchaud fractional derivative on the real line (Definition 6.1.18)

Representations of FBM are obtained in terms of fractional integrals and derivatives in Section 6.2. These representations then play a central role in developing deterministic integration with respect to FBM, and in being able to address various applications involving that process.

Integrals of *deterministic* functions with respect to FBM will be called *fractional Wiener integrals*. They are introduced in Section 6.3 and defined first as integrals on an interval (Section 6.3.2) and then as integrals on the real line (Section 6.3.6). Applications involving the Girsanov formula, prediction and filtering can be found in Section 6.4. Integrals of *random* functions with respect to FBM are introduced in Chapter 7.

6.1 Fractional Integrals and Derivatives

We provide here the basics of fractional integrals and derivatives on an interval (Sections 6.1.1 and 6.1.2) and fractional integrals and derivatives on the real line (Sections 6.1.3–6.1.5).

6.1.1 Fractional Integrals on an Interval

The standard way of motivating the definition of a fractional integral is to start with an n -tuple iterated integral and show that it can be expressed as a single integral involving the parameter n . The fractional integral of order $\alpha > 0$ is then defined by replacing the integer n by the real number α in the resulting expression.

Let $a < b$ be two real numbers and ϕ be an integrable function on $[a, b]$. A multiple integral of ϕ can be expressed as

$$\int_a^{t_n} \dots \left\{ \int_a^{t_2} \left\{ \int_a^{t_1} \phi(u) du \right\} dt_1 \right\} \dots dt_{n-1} = \frac{1}{(n-1)!} \int_a^{t_n} \phi(u)(t_n - u)^{n-1} du, \quad (6.1.1)$$

where $t_n \in [a, b]$ and $n \geq 1$. (By convention, $(0)! = 1$ and $a^0 = 1$.) The proof is by induction. Consider $n = 2$. We have $a < u < t_1 < t_2$. Then, by changing the order of integration, we get

$$\int_a^{t_2} \int_a^{t_1} \phi(u) du dt_1 = \int_a^{t_2} \phi(u) \int_u^{t_2} dt_1 du = \int_a^{t_2} \phi(u)(t_2 - u) du,$$

so that the relation (6.1.1) holds for $n = 2$. Suppose now that it holds for $n - 1$ and consider it for n . By induction and the change of the order of integration, we get in the same way that

$$\begin{aligned} \int_a^{t_n} \dots \int_a^{t_2} \int_a^{t_1} \phi(u) du dt_1 \dots dt_{n-1} &= \frac{1}{(n-2)!} \int_a^{t_n} \int_a^{t_{n-1}} \phi(u)(t_{n-1} - u)^{n-2} du dt_{n-1} \\ &= \frac{1}{(n-2)!} \int_a^{t_n} \phi(u) \int_u^{t_{n-1}} (t_{n-1} - u)^{n-2} dt_{n-1} du = \frac{1}{(n-1)!} \int_a^{t_n} \phi(u)(t_n - u)^{n-1} du, \end{aligned}$$

which is the relation (6.1.1). The idea behind fractional integrals is to replace an integer n in (6.1.1) by a real number $\alpha > 0$ (hence the name “fractional integrals”). Observe also that $(n-1)! = \Gamma(n)$, where Γ is the gamma function. We write $L^p[a, b]$ for the space $L^p([a, b], du)$, $p > 0$.

Definition 6.1.1 Let $\phi \in L^1[a, b]$ and $\alpha > 0$. The integrals

$$(I_{a+}^\alpha \phi)(s) = \frac{1}{\Gamma(\alpha)} \int_a^s \phi(u)(s-u)^{\alpha-1} du = \frac{1}{\Gamma(\alpha)} \int_a^b \phi(u)(s-u)_+^{\alpha-1} du, \quad (6.1.2)$$

$s \in (a, b)$, and

$$(I_{b-}^\alpha \phi)(s) = \frac{1}{\Gamma(\alpha)} \int_s^b \phi(u)(u-s)^{\alpha-1} du = \frac{1}{\Gamma(\alpha)} \int_a^b \phi(u)(u-s)_+^{\alpha-1} du, \quad (6.1.3)$$

$s \in (a, b)$, are called *fractional integrals* of order α . (For any real number $x \neq 0$, $x^0 = 1$.) The fractional integral I_{a+}^α is called *left-sided* since the integration in (6.1.2) is over $[a, s]$ which is the left side of the interval $[a, b]$. The integral I_{b-}^α is called *right-sided* since the integration in (6.1.3) is over $[s, b]$. Both (6.1.2) and (6.1.3) are also called *Riemann–Liouville fractional integrals*.

Note that the integrals $I_{a+}^\alpha \phi$ and $I_{b-}^\alpha \phi$ are well-defined a.e. for functions $\phi \in L^1[a, b]$ (and hence, for functions $\phi \in L^p[a, b]$, $p > 1$, as well). Indeed, for the integral $I_{a+}^\alpha \phi$, we have

$$\begin{aligned} \Gamma(\alpha) \int_a^b (I_{a+}^\alpha |\phi|)(s) ds &= \int_a^b \int_a^b |\phi(u)|(s-u)_+^{\alpha-1} du ds = \int_a^b du |\phi(u)| \int_a^b ds (s-u)_+^{\alpha-1} \\ &= \alpha^{-1} \int_a^b |\phi(u)|(b-u)^\alpha du \leq \alpha^{-1} (b-a)^\alpha \int_a^b |\phi(u)| du < \infty, \end{aligned}$$

so that $(I_{a+}^\alpha |\phi|)(s) < \infty$ and hence $(I_{a+}^\alpha \phi)(s)$ is well-defined ds -a.e. for $s \in (a, b)$. The case of the integral $I_{b-}^\alpha \phi$ is similar. Note also that, if $\alpha \geq 1$ and $\phi \in L^1[a, b]$, the integrals

$I_{a+}^\alpha \phi$ and $I_{b-}^\alpha \phi$ are well-defined at every point $s \in (a, b)$. For example, for $I_{a+}^\alpha \phi$, this follows by bounding $(s - u)^{\alpha-1}$ by $(b - a)^{\alpha-1}$.

Example 6.1.2 (Fractional integrals of power functions) Let $\phi(u) = (u - a)^{-\beta}$, for $\beta < 1$. Then we have, after the change of variables $u = a + (s - a)z$,

$$\begin{aligned} (I_{a+}^\alpha \phi)(s) &= \frac{1}{\Gamma(\alpha)} \int_a^s (u - a)^{-\beta} (s - u)^{\alpha-1} du = \frac{(s - a)^{\alpha-\beta}}{\Gamma(\alpha)} \int_0^1 z^{-\beta} (1 - z)^{\alpha-1} dz \\ &= \frac{B(1 - \beta, \alpha)}{\Gamma(\alpha)} (s - a)^{\alpha-\beta} = \frac{\Gamma(1 - \beta)}{\Gamma(1 + \alpha - \beta)} (s - a)^{\alpha-\beta}, \end{aligned}$$

where B is the beta function defined in (2.2.12). The exponent in the function ϕ has increased by α .

The following result gathers some basic properties of fractional integrals.

Proposition 6.1.3 Let $\alpha > 0$. The fractional integrals I_{a+}^α and I_{b-}^α have the following properties:

(i) *Reflection property*: if Q is the “reflection operator” defined by $(Q\phi)(u) = \phi(a + b - u)$ for $u \in [a, b]$, then

$$QI_{a+}^\alpha = I_{b-}^\alpha Q \quad \text{and} \quad QI_{b-}^\alpha = I_{a+}^\alpha Q. \quad (6.1.4)$$

(ii) *Semigroup property*: for $\phi \in L^1[a, b]$,

$$I_{a+}^\alpha I_{a+}^\beta \phi = I_{a+}^{\alpha+\beta} \phi, \quad I_{b-}^\alpha I_{b-}^\beta \phi = I_{b-}^{\alpha+\beta} \phi, \quad \alpha, \beta > 0. \quad (6.1.5)$$

The equations (6.1.5) hold at almost every point. They also hold at every point if $\alpha + \beta \geq 1$.

(iii) *Fractional integration by parts formula*: suppose $\phi \in L^p[a, b]$ and $\psi \in L^q[a, b]$ either with $\alpha \geq 1$, $p = 1$, $q = 1$, or with $0 < \alpha < 1$, $1/p + 1/q \leq 1 + \alpha$, $p > 1$, $q > 1$. Then

$$\int_a^b \phi(s) (I_{a+}^\alpha \psi)(s) ds = \int_a^b (I_{b-}^\alpha \phi)(s) \psi(s) ds. \quad (6.1.6)$$

Proof The property (i) is elementary since by setting $v = Qu = a + b - u$,

$$\int_a^{a+b-s} \phi(u) (a + b - s - u)^{\alpha-1} du = \int_s^b \phi(a + b - v) (v - s)^{\alpha-1} dv$$

(see Figure 6.1). To show the first relation in the property (ii), note that

$$(I_{a+}^\alpha I_{a+}^\beta \phi)(s) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_a^s (s - u)^{\alpha-1} \int_a^u \phi(v) (u - v)^{\beta-1} dv du.$$



Figure 6.1 The reflection Q moves the argument u into Qu such that $u - a = b - Qu$.

By changing the order of integration and making the change of variables $u = v + (s - v)w$,

$$\begin{aligned}(I_{a+}^{\alpha} I_{a+}^{\beta} \phi)(s) &= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_a^s \phi(v) \int_v^s (s-u)^{\alpha-1} (u-v)^{\beta-1} dudv \\ &= \frac{B(\alpha, \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_a^s \phi(v) (s-v)^{\alpha+\beta-1} dv = (I_{a+}^{\alpha+\beta} \phi)(s),\end{aligned}$$

where $B(\alpha, \beta)$ is the beta function defined in (2.2.12). The change of variables is valid and the relation holds under the assumptions made – see the discussion following Definition 6.1.1. The second relation in (6.1.5) can be proved similarly.

The property (iii) would follow immediately if one could change the order of integration in:

$$\begin{aligned}\int_a^b \phi(s) (I_{a+}^{\alpha} \psi)(s) ds &= \frac{1}{\Gamma(\alpha)} \int_a^b \phi(s) \int_a^s \psi(u) (s-u)^{\alpha-1} dud s \\ &= \frac{1}{\Gamma(\alpha)} \int_a^b \psi(u) \int_u^b \phi(s) (s-u)^{\alpha-1} ds du = \int_a^b (I_{b-}^{\alpha} \phi)(s) \psi(s) ds.\end{aligned}$$

Showing the validity of changing the order of integration (by Fubini's theorem) is more delicate. It is necessary to understand the spaces of functions to which functions are mapped by fractional integral operators. See Theorem 3.5 and its Corollary in Samko, Kilbas, and Marichev [878]. $\square$

6.1.2 Riemann–Liouville Fractional Derivatives $\mathcal{D}$ on an Interval

To obtain fractional derivatives, one needs to extend the definition of fractional integrals to $\alpha < 0$. Observe, however, that we can not simply substitute $\alpha < 0$ into (6.1.2) or (6.1.3) since the resulting integrals diverge. As is the case in ordinary calculus, the definition of a derivative involves more restrictions than that of an integral. We shall suppose first that $-1 < \alpha < 0$. The operator $I_{a+}^{-\alpha}$ for $0 < \alpha < 1$ is introduced as the inverse of I_{a+}^{α} . In other words, consider the equation

$$(I_{a+}^{\alpha} \phi)(s) = \frac{1}{\Gamma(\alpha)} \int_a^s \phi(z) (s-z)^{\alpha-1} dz = f(s), \quad s \in (a, b), \quad 0 < \alpha < 1, \quad (6.1.7)$$

where $\phi \in L^1[a, b]$. (The equation (6.1.7) is called *Abel's equation*.) We want to solve it for functions ϕ assuming f is well-behaved. Multiplying both sides of (6.1.7) by $(u-s)^{-\alpha}$, where $u \in (a, b)$, and integrating we get

$$\int_a^u (u-s)^{-\alpha} \int_a^s \phi(z) (s-z)^{\alpha-1} dz ds = \Gamma(\alpha) \int_a^u f(s) (u-s)^{-\alpha} ds. \quad (6.1.8)$$

Changing the order of integration

$$\int_a^u \phi(z) \int_z^u (u-s)^{-\alpha} (s-z)^{\alpha-1} ds dz = \Gamma(\alpha) \int_a^u f(s) (u-s)^{-\alpha} ds$$

and using the fact

$$\int_z^u (u-s)^{-\alpha} (s-z)^{\alpha-1} ds = B(\alpha, 1-\alpha) = \Gamma(\alpha)\Gamma(1-\alpha), \quad (6.1.9)$$

which follows from (2.2.12), we obtain that

$$\int_a^u \phi(z) dz = \frac{1}{\Gamma(1-\alpha)} \int_a^u f(s)(u-s)^{-\alpha} ds. \quad (6.1.10)$$

Now, differentiating both sides with respect to u yields

$$\phi(u) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_a^u f(s)(u-s)^{-\alpha} ds.$$

The operator $I_{a+}^{-\alpha}$ for $0 < \alpha < 1$ is then defined as follows.

Definition 6.1.4 Let $0 < \alpha < 1$. The integrals

$$(\mathcal{D}_{a+}^{\alpha} f)(u) = (I_{a+}^{-\alpha} f)(u) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_a^u f(s)(u-s)^{-\alpha} ds, \quad (6.1.11)$$

$u \in (a, b)$, and

$$(\mathcal{D}_{b-}^{\alpha} f)(u) = (I_{b-}^{-\alpha} f)(u) = -\frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_u^b f(s)(s-u)^{-\alpha} ds, \quad (6.1.12)$$

$u \in (a, b)$, are called *fractional derivatives* of order α . The fractional derivatives $\mathcal{D}_{a+}^{\alpha}$ and $\mathcal{D}_{b-}^{\alpha}$ are called left-sided and right-sided, respectively. Both (6.1.11) and (6.1.12) are also called the *Riemann–Liouville fractional derivatives*.

Remark 6.1.5 In contrast to the usual derivative, the fractional derivative is not a local operation since $(\mathcal{D}_{a+}^{\alpha} f)(u)$ does not depend only on the behavior of f in a neighborhood of u .

Remark 6.1.6 Note that, heuristically, by taking the derivative d/du inside (6.1.11), we get

$$\frac{-\alpha}{\Gamma(1-\alpha)} \int_a^u f(s)(u-s)^{-\alpha-1} ds = \frac{1}{\Gamma(-\alpha)} \int_a^u f(s)(u-s)^{-\alpha-1} ds,$$

which is exactly the fractional integral (6.1.2) with $-1 < -\alpha < 0$. Doing the same thing inside (6.1.12), we get (6.1.3), explaining the presence of the additional minus sign in (6.1.12).

The fractional derivatives $\mathcal{D}_{a+}^{\alpha} f$ and $\mathcal{D}_{b-}^{\alpha} f$ are well-defined if, for example, f is an absolutely continuous function; that is, $f(s) = f(a) + \int_a^s f'(t) dt$ for $s \in [a, b]$. Indeed, we have¹

¹ It is useful to recall the formula for $\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt$. Assume that the functions $f(x, t)$ and $\frac{\partial}{\partial x} f(x, t)$ are continuous in the region $x_0 \leq x \leq x_1$, $a(x) \leq t \leq b(x)$ of the (x, t) plane, and that $a(x)$ and $b(x)$ are both continuous with continuous derivatives for $x_0 \leq x \leq x_1$. Then, for any $x_0 < x < x_1$,

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, t) dt = f(x, b(x))b'(x) - f(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt.$$

This formula is known as Leibniz's rule for differentiation under the integral sign.

$$\begin{aligned}
(\mathcal{D}_{a+}^\alpha f)(s) &= \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_a^u \left(f(a) + \int_a^s f'(t) dt \right) (u-s)^{-\alpha} ds \\
&= \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \left(f(a) \int_a^u (u-s)^{-\alpha} ds + \int_a^u dt f'(t) \int_t^u ds (u-s)^{-\alpha} \right) \\
&= \frac{1}{(1-\alpha)\Gamma(1-\alpha)} \frac{d}{du} \left(f(a)(u-a)^{1-\alpha} + \int_a^u f'(t)(u-t)^{1-\alpha} dt \right) \\
&= \frac{1}{\Gamma(1-\alpha)} \left(f(a)(u-a)^{-\alpha} + \int_a^u f'(t)(u-t)^{-\alpha} dt \right).
\end{aligned}$$

The case of $\mathcal{D}_{b-}^\alpha f$ is similar. These results should not be surprising. We indeed expect that if f is differentiable then it also should be fractionally differentiable with order $0 < \alpha < 1$.

Example 6.1.7 (Fractional derivatives of power functions) Let $0 < \alpha < 1$ and $f(s) = (s-a)^{\beta-1}$ with $s \in (a, b)$ and $\beta > 0$. Then we have that after the change of variables $s = a + (u-a)z$ and (2.2.12),

$$\begin{aligned}
(\mathcal{D}_{a+}^\alpha f)(u) &= \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_a^u (s-a)^{\beta-1} (u-s)^{-\alpha} ds = \frac{B(\beta, 1-\alpha)}{\Gamma(1-\alpha)} \frac{d}{du} (u-a)^{\beta-\alpha} \\
&= \frac{\Gamma(\beta)}{\Gamma(1+\beta-\alpha)} \begin{cases} (\beta-\alpha)(u-a)^{\beta-\alpha-1}, & \text{if } \beta \neq \alpha, \\ 0, & \text{if } \beta = \alpha \end{cases} = \begin{cases} \frac{\Gamma(\beta)}{\Gamma(\beta-\alpha)} (u-a)^{\beta-\alpha-1}, & \text{if } \beta \neq \alpha, \\ 0, & \text{if } \beta = \alpha. \end{cases}
\end{aligned}$$

Observe that if $\beta \neq \alpha$, then the exponent $\beta-1$ in the function f decreases by α . Now suppose that $\beta = 1$, so that $f(s) \equiv 1$ is constant. Its fractional derivative is not zero if $0 < \alpha < 1$, but is equal to

$$(\mathcal{D}_{a+}^\alpha 1)(u) = \frac{(u-a)^{-\alpha}}{\Gamma(1-\alpha)}.$$

The next result provides basic properties of fractional derivatives. We consider only $\mathcal{D}_{a+}^\alpha$. Similar relations hold for $\mathcal{D}_{b-}^\alpha$.

Proposition 6.1.8 Let $0 < \alpha < 1$. The fractional derivative $\mathcal{D}_{a+}^\alpha$ has the following properties:

(i) for any $\phi \in L^1[a, b]$, we have that

$$\mathcal{D}_{a+}^\alpha I_{a+}^\alpha \phi = \phi. \quad (6.1.13)$$

(ii) for any f such that $f = I_{a+}^\alpha \phi$, where $\phi \in L^1[a, b]$, we have that

$$I_{a+}^\alpha \mathcal{D}_{a+}^\alpha f = f. \quad (6.1.14)$$

(iii) if the function $I_{a+}^{1-\alpha} f$ is absolutely continuous, then

$$(I_{a+}^\alpha \mathcal{D}_{a+}^\alpha f)(s) = f(s) - \frac{(I_{a+}^{1-\alpha} f)(a)}{\Gamma(\alpha)} (s-a)^{\alpha-1}, \quad s \in (a, b), \quad (6.1.15)$$

where $(I_{a+}^{1-\alpha} f)(a) = \lim_{s \downarrow a} (I_{a+}^{1-\alpha} f)(s)$.

Remark 6.1.9 In (ii) and (iii) we have special conditions on f . This makes sense since we differentiate first before integrating.

Remark 6.1.10 The last term in (6.1.15) can be thought as the boundary term resulting from the integration I_{a+}^{α} . Its presence is naturally expected in analogy to the fundamental calculus relation $(I_{a+}^1 \mathcal{D}_{a+}^1 f)(s) = \int_a^s f'(t)dt = f(s) - f(a)$. This term is zero in (6.1.14) since $(I_{a+}^{1-\alpha} f)(a) = (I_{a+}^{1-\alpha} I_{a+}^{\alpha} \phi)(a) = (I_{a+}^1 \phi)(a) = \int_a^a \phi(u)du = 0$.

Proof For the property (i), note that, by arguing as in (6.1.8)–(6.1.10),

$$\begin{aligned} (\mathcal{D}_{a+}^{\alpha} I_{a+}^{\alpha} \phi)(u) &= \frac{1}{\Gamma(1-\alpha)\Gamma(\alpha)} \frac{d}{du} \int_a^u (u-s)^{-\alpha} \int_a^s \phi(v)(s-v)^{\alpha-1} dv ds \\ &= \frac{d}{du} \int_a^u \phi(v)dv = \phi(u). \end{aligned}$$

The property (ii) follows immediately from (i). For the property (iii), note first that, by the assumption that $I_{a+}^{1-\alpha} f$ is absolutely continuous, $(I_{a+}^{1-\alpha} f)(s) = (I_{a+}^1 g)(s) + (I_{a+}^{1-\alpha} f)(a)$ for some $g \in L^1[a, b]$. By using Example 6.1.2, note that $(I_{a+}^{1-\alpha} (u-a)^{\alpha-1})(s) = \frac{\Gamma(\alpha)}{\Gamma(1)}(s-a)^{1-\alpha+\alpha-1} = \Gamma(\alpha)$. Hence, we can also write

$$\left(I_{a+}^{1-\alpha} \left(f(u) - \frac{(I_{a+}^{1-\alpha} f)(a)}{\Gamma(\alpha)} (u-a)^{\alpha-1} \right) \right)(s) = (I_{a+}^1 g)(s).$$

By applying $\mathcal{D}_{a+}^{1-\alpha}$ to both sides, and using the property (ii) above and the property (ii) of Proposition 6.1.3, we obtain

$$f(u) - \frac{(I_{a+}^{1-\alpha} f)(a)}{\Gamma(\alpha)} (u-a)^{\alpha-1} = (\mathcal{D}_{a+}^{1-\alpha} I_{a+}^1 g)(u) = (\mathcal{D}_{a+}^{1-\alpha} I_{a+}^{1-\alpha} I_{a+}^{\alpha} g)(u) = (I_{a+}^{\alpha} g)(u).$$

Then, by the property (ii),

$$\left(I_{a+}^{\alpha} \mathcal{D}_{a+}^{\alpha} \left(f(u) - \frac{(I_{a+}^{1-\alpha} f)(a)}{\Gamma(\alpha)} (u-a)^{\alpha-1} \right) \right)(s) = f(s) - \frac{(I_{a+}^{1-\alpha} f)(a)}{\Gamma(\alpha)} (s-a)^{\alpha-1}.$$

The relation (6.1.15) now follows by noting that $(\mathcal{D}_{a+}^{\alpha} (u-a)^{\alpha-1})(z) = 0$ by Example 6.1.7. $\square$

Corollary 6.1.11 For $\alpha > 0$, $I_{a+}^{\alpha} \phi = 0$ or $I_{b-}^{\alpha} \phi = 0$ implies that $\phi = 0$ almost everywhere.

Proof We consider only $I_{a+}^{\alpha} \phi$. When $0 < \alpha < 1$, the result follows from Proposition 6.1.8, (i), since $\phi = \mathcal{D}_{a+}^{\alpha} I_{a+}^{\alpha} \phi = 0$ a.e. When $\alpha \geq 1$, $I_{a+}^{\alpha} \phi = 0$ implies $I_{a+}^{[\alpha]} I_{a+}^{\alpha-[\alpha]} \phi = 0$, where $[\alpha]$ is the integer part of α . Using the relation (6.1.1) and differentiating the last expression $[\alpha]$ number of times leads to $I_{a+}^{\alpha-[\alpha]} \phi = 0$. Since $0 \leq \alpha - [\alpha] < 1$, we also get $\phi = 0$ a.e. $\square$

One may define the fractional derivative operators $\mathcal{D}_{a+}^{\alpha}$ and $\mathcal{D}_{b-}^{\alpha}$ for $\alpha \geq 1$ as well. The idea is to use the usual integer order derivative operators. For example, $(\mathcal{D}_{a+}^{1.7} f)(u)$

would be defined as $\frac{d}{du}((\mathcal{D}_{a+}^{0.7}f)(u)) = \frac{d^2}{du^2}((I_{a+}^{0.3}f)(u))$, where the equality follows from the definitions (6.1.11) and (6.1.2). Because we are interested in fractional Brownian motion we will focus on the range $\alpha \in (-1/2, 1/2)$ only.

Remark 6.1.12 Set $a = 0$. As indicated above, e.g., the Riemann–Liouville derivative $\mathcal{D}_{0+}^\alpha$ for $\alpha > 0$ is defined as $\frac{d^n}{du^n} I_{0+}^{n-\alpha}$ where $n = [\alpha] + 1$. If the order of the last two operations is reversed, that is, $I_{0+}^{n-\alpha} \frac{d^n}{du^n}$ is considered, one gets the so-called Caputo derivative. Note that while the Caputo derivative of a constant is always zero, this is not generally the case for the Riemann–Liouville derivative (as can be seen from Example 6.1.7 with $\beta = 1$). For more information on the Caputo derivative, see Podlubny [825], Meerschaert and Siko-rskii [712]. In the context of FBM, the Riemann–Liouville derivative is sufficient for our purposes.

6.1.3 Fractional Integrals and Derivatives on the Real Line

Fractional integrals and derivatives can also be defined on the real line. Two types of fractional derivatives will be defined: the Liouville fractional derivatives and the Marchaud fractional derivatives.

Definition 6.1.13 Let $\alpha > 0$. The integrals

$$\begin{aligned} (I_+^\alpha \phi)(s) &= \frac{1}{\Gamma(\alpha)} \int_{-\infty}^s \phi(u)(s-u)^{\alpha-1} du = \frac{1}{\Gamma(\alpha)} \int_{\mathbb{R}} \phi(u)(s-u)_+^{\alpha-1} du \\ &= \frac{1}{\Gamma(\alpha)} \int_0^\infty \phi(s-y)y^{\alpha-1} dy, \end{aligned} \quad (6.1.16)$$

$s \in \mathbb{R}$, and

$$\begin{aligned} (I_-^\alpha \phi)(s) &= \frac{1}{\Gamma(\alpha)} \int_s^\infty \phi(u)(u-s)^{\alpha-1} du = \frac{1}{\Gamma(\alpha)} \int_{\mathbb{R}} \phi(u)(u-s)_+^{\alpha-1} du \\ &= \frac{1}{\Gamma(\alpha)} \int_0^\infty \phi(s+y)y^{\alpha-1} dy, \end{aligned} \quad (6.1.17)$$

$s \in \mathbb{R}$, are called *fractional integrals* of order α on the real line.

Definition 6.1.14 Let $0 < \alpha < 1$. The integrals

$$(\mathcal{D}_+^\alpha f)(u) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_{-\infty}^u f(s)(u-s)^{-\alpha} ds = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_0^\infty f(u-y)y^{-\alpha} dy, \quad (6.1.18)$$

$u \in \mathbb{R}$, and

$$(\mathcal{D}_-^\alpha f)(u) = -\frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_u^\infty f(s)(s-u)^{-\alpha} ds = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_0^\infty f(u+y)y^{-\alpha} dy, \quad (6.1.19)$$

$u \in \mathbb{R}$, are called *Liouville fractional derivatives* of order α on the real line.

Remark 6.1.15 Note that these definitions are similar to Definition 6.1.1 and Definition 6.1.4, respectively, but with $a = -\infty$ and $b = \infty$. The basic question, however, is: *What is the domain of these new fractional operators?*

What about the domain of the operators $I_{\pm}^{\alpha}$? Let us observe that $I_{\pm}^{\alpha}\phi$ is well-defined for $0 < \alpha < 1$ and $\phi \in L^p(\mathbb{R})$ with $1 \leq p < 1/\alpha$. Indeed, we have

$$\Gamma(\alpha)(I_{+}^{\alpha}\phi)(s) = \int_0^1 \phi(s-v)v^{\alpha-1}dv + \int_1^{\infty} \phi(s-v)v^{\alpha-1}dv. \quad (6.1.20)$$

The second term is bounded for all s by Hölder's inequality when $1 < p < 1/\alpha$ and simply by $\|\phi\|_{L^1(\mathbb{R})}$ when $p = 1$. The first term is bounded by the generalized Minkowski's inequality. Recall that the generalized Minkowski's inequality (e.g., Zygmund [1030], p. 19), given by

$$\left\{ \int_{\Omega_1} dx \left| \int_{\Omega_2} f(x, y) dy \right|^p \right\}^{1/p} \leq \int_{\Omega_2} dy \left\{ \int_{\Omega_1} |f(x, y)|^p dx \right\}^{1/p}, \quad (6.1.21)$$

is valid for all $p \geq 1$ and any (possibly infinite) intervals Ω_1 and Ω_2 . If $\Omega_2 = (0, 2)$ and $f(x, y) = f(x)$ for $0 < y < 1$ and $f(x, y) = g(x)$ for $1 < y < 2$, then one obtains the usual Minkowski's inequality $\|f + g\|_{L^p(\Omega_1)} \leq \|f\|_{L^p(\Omega_1)} + \|g\|_{L^p(\Omega_1)}$. To show that the first term of (6.1.20) is bounded, apply (6.1.21) with $\Omega_1 = \mathbb{R}$ and $\Omega_2 = [0, 1]$ to write

$$\begin{aligned} \left\{ \int_{\mathbb{R}} ds \left| \int_0^1 |\phi(s-v)v^{\alpha-1}dv| \right|^p \right\}^{1/p} &\leq \int_0^1 dv \left\{ \int_{\mathbb{R}} |\phi(s-v)v^{\alpha-1}|^p ds \right\}^{1/p} \\ &= \int_0^1 v^{\alpha-1} dv \|\phi\|_{L^p(\mathbb{R})} < \infty. \end{aligned}$$

This yields $\int_0^1 |\phi(s-v)v^{\alpha-1}dv| < \infty$ a.e. ds and hence that $\int_0^1 \phi(s-v)v^{\alpha-1}dv$ is well-defined a.e. ds .

Example 6.1.16 (*Fractional integrals and derivatives of an exponential function*) Consider the exponential function $\phi(u) = e^{\lambda u}$ with parameter $\lambda > 0$. Then, by (6.1.16) for $\alpha > 0$,

$$(I_{+}^{\alpha}\phi)(s) = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} e^{\lambda(s-y)} y^{\alpha-1} dy = \frac{e^{\lambda s}}{\Gamma(\alpha)} \int_0^{\infty} e^{-\lambda y} y^{\alpha-1} dy = e^{\lambda s} \lambda^{-\alpha}, \quad s \in \mathbb{R}, \quad (6.1.22)$$

and by (6.1.18) for $0 < \alpha < 1$,

$$(\mathcal{D}_{+}^{\alpha}\phi)(u) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_0^{\infty} e^{\lambda(u-y)} y^{-\alpha} dy = \frac{d}{du} e^{\lambda u} \lambda^{\alpha-1} = e^{\lambda u} \lambda^{\alpha}, \quad u \in \mathbb{R}. \quad (6.1.23)$$

Note that $I_{-}^{\alpha}\phi$ and $\mathcal{D}_{-}^{\alpha}\phi$ are not considered since the integrals entering their definitions are infinite.

Example 6.1.17 (*Fractional integrals and derivatives of indicator functions*) We want to fractionally integrate and differentiate an indicator function. Let $a < b$ be real numbers. Then, for $\alpha > 0$,

$$\begin{aligned}\Gamma(1+\alpha)(I_{\pm}^{\alpha}1_{[a,b]})(s) &= \frac{\Gamma(1+\alpha)}{\Gamma(\alpha)} \int_{\mathbb{R}} 1_{[a,b]}(u)(u-s)_{\mp}^{\alpha-1} du \\ &= \alpha \int_a^b (u-s)_{\mp}^{\alpha-1} du = (b-s)_{\mp}^{\alpha} - (a-s)_{\mp}^{\alpha}, \quad s \in \mathbb{R}. \quad (6.1.24)\end{aligned}$$

Similarly, for $0 < \alpha < 1$, we have that

$$\begin{aligned}\Gamma(1-\alpha)(\mathcal{D}_{\pm}^{\alpha}1_{[a,b]})(u) &= \pm \frac{d}{du} \int_{\mathbb{R}} 1_{[a,b]}(s)(s-u)_{\mp}^{-\alpha} ds \\ &= \pm \frac{1}{1-\alpha} \frac{d}{du} ((b-u)_{\mp}^{1-\alpha} - (a-u)_{\mp}^{1-\alpha}) = (b-u)_{\mp}^{-\alpha} - (a-u)_{\mp}^{-\alpha}, \quad u \in \mathbb{R}. \quad (6.1.25)\end{aligned}$$

The fractional integrals $I_{\pm}^{\alpha}$ have properties analogous to those of fractional integrals on an interval in Proposition 6.1.3. So, for example, we have $QI_{\pm}^{\alpha}\phi = I_{\mp}^{\alpha}Q\phi$, where the “reflection operator” Q is defined by $(Q\phi)(u) = \phi(-u)$ for $u \in \mathbb{R}$. We similarly have that, for “sufficiently good” functions ϕ and ψ , the *semigroup property* (Samko et al. [878], p. 34)

$$I_{\pm}^{\alpha}I_{\pm}^{\beta}\phi = I_{\pm}^{\alpha+\beta}\phi, \quad \alpha, \beta > 0, \quad (6.1.26)$$

and the *fractional integration by parts formula* (Samko et al. [878], p. 34)

$$\int_{\mathbb{R}} \phi(s)(I_{+}^{\alpha}\psi)(s)ds = \int_{\mathbb{R}} (I_{-}^{\alpha}\phi)(s)\psi(s)ds \quad (6.1.27)$$

hold. If $I_{\pm}^{\alpha}\phi = 0$, we also expect that $\phi = 0$ almost everywhere (cf. Corollary 6.1.11).

What about properties of fractional derivatives $\mathcal{D}_{\pm}^{\alpha}$? For example, do we have $\mathcal{D}_{\pm}^{\alpha}I_{\pm}^{\alpha}\phi = \phi$? The answer is yes, but only for functions $\phi \in L^1(\mathbb{R})$. (Indeed, as in the proof of property (i) of Proposition 6.1.8 we have that $\mathcal{D}_{+}^{\alpha}I_{+}^{\alpha}\phi = \frac{d}{du} \int_{-\infty}^u \phi(s)ds$ which assumes summability of the function ϕ at infinity.) It turns out that this property is not strong enough in the context of fractional Brownian motion. There is, however, another fractional derivative which allows for greater flexibility.

6.1.4 Marchaud Fractional Derivatives D on the Real Line

Observe that for the Liouville fractional derivative $\mathcal{D}_{+}^{\alpha}f$ with a “sufficiently good” function f , we have

$$\begin{aligned}(\mathcal{D}_{+}^{\alpha}f)(u) &= \frac{1}{\Gamma(1-\alpha)} \frac{d}{du} \int_0^{\infty} v^{-\alpha} f(u-v)dv = \frac{1}{\Gamma(1-\alpha)} \int_0^{\infty} v^{-\alpha} f'(u-v)dv \\ &= \frac{\alpha}{\Gamma(1-\alpha)} \int_0^{\infty} f'(u-v)dv \int_v^{\infty} \frac{d\xi}{\xi^{1+\alpha}} = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^{\infty} \frac{d\xi}{\xi^{1+\alpha}} \int_0^{\xi} f'(u-v)dv \\ &= \frac{\alpha}{\Gamma(1-\alpha)} \int_0^{\infty} \frac{f(u) - f(u-\xi)}{\xi^{1+\alpha}} d\xi. \quad (6.1.28)\end{aligned}$$

The corresponding relation for $\mathcal{D}_{-}^{\alpha}f$ is given by

$$(\mathcal{D}_{-}^{\alpha}f)(u) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^{\infty} \frac{f(u) - f(u+\xi)}{\xi^{1+\alpha}} d\xi. \quad (6.1.29)$$

It is clear that the operators $\mathcal{D}_{\pm}^{\alpha}$ in Definition 6.1.14 and the operators defined by the right-hand side of (6.1.28) and (6.1.29) have different domains. So, for example, we may take f to behave like a constant at infinity or even grow slowly enough at infinity for the right-hand side of (6.1.28) or (6.1.29) to exist whereas $\mathcal{D}_{\pm}^{\alpha}f$ is not defined for such f since constants are not in $L^1(\mathbb{R})$.

Definition 6.1.18 Let $\alpha \in (0, 1)$. The operators

$$\mathbf{D}_{\pm}^{\alpha}f = \lim_{\epsilon \rightarrow 0} \mathbf{D}_{\pm, \epsilon}^{\alpha}f, \quad (6.1.30)$$

where

$$(\mathbf{D}_{\pm, \epsilon}^{\alpha}f)(u) = \frac{\alpha}{\Gamma(1-\alpha)} \int_{\epsilon}^{\infty} \frac{f(u) - f(u \mp \xi)}{\xi^{1+\alpha}} d\xi, \quad (6.1.31)$$

are called *Marchaud fractional derivatives* on the real line of order α . The expressions in (6.1.31) are called *truncated Marchaud fractional derivatives*.

Remark 6.1.19 There is another way to obtain Marchaud fractional derivatives. Let $-1 < \alpha < 0$. Heuristically, by substituting this negative α into the expression (6.1.17) of I_{-}^{α} we get that

$$(I_{-}^{\alpha}\phi)(s) = \frac{1}{\Gamma(-|\alpha|)} \int_0^{\infty} \phi(v+s)v^{-|\alpha|-1}dv. \quad (6.1.32)$$

If ϕ is not trivial, this integral diverges at $v = 0$ like

$$\frac{1}{\Gamma(-|\alpha|)} \int_0^{\infty} \phi(s)v^{-|\alpha|-1}dv. \quad (6.1.33)$$

But by subtracting (6.1.33) from (6.1.32), we get

$$\begin{aligned} \frac{1}{\Gamma(-|\alpha|)} \int_0^{\infty} (\phi(v+s) - \phi(s))v^{-|\alpha|-1}dv &= \frac{-|\alpha|}{\Gamma(1-|\alpha|)} \int_0^{\infty} (\phi(v+s) - \phi(s))v^{-|\alpha|-1}dv \\ &= \frac{|\alpha|}{\Gamma(1-|\alpha|)} \int_0^{\infty} (\phi(s) - \phi(v+s))v^{-|\alpha|-1}dv \\ &= (\mathbf{D}_{-}^{-\alpha}\phi)(s), \end{aligned}$$

since $|\alpha|\Gamma(-|\alpha|) = -\Gamma(1-|\alpha|)$.

We have used a “renormalization” perspective by compensating for the divergence. Let us explain what we did in more rigorous terms. For $\epsilon > 0$, write

$$\frac{1}{\Gamma(-|\alpha|)} \int_{\epsilon}^{\infty} \phi(v+s)v^{-|\alpha|-1}dv = \frac{\phi(s+\epsilon)\epsilon^{-|\alpha|}}{|\alpha|\Gamma(-|\alpha|)} + J_{\epsilon},$$

where, using $|\alpha|\Gamma(-|\alpha|) = -\Gamma(1-|\alpha|)$,

$$J_{\epsilon} = \frac{\phi(s+\epsilon)\epsilon^{-|\alpha|}}{\Gamma(1-|\alpha|)} - \frac{|\alpha|}{\Gamma(1-|\alpha|)} \int_{\epsilon}^{\infty} \phi(v+s)v^{-|\alpha|-1}dv.$$

(The term $\phi(s+\epsilon)\epsilon^{-|\alpha|}$ arises as the boundary terms at $v = \epsilon$ in the integration by parts of $-|\alpha| \int_{\epsilon}^{\infty} \phi(v+s)v^{-|\alpha|-1}dv = \int_{\epsilon}^{\infty} \phi(v+s)dv^{-|\alpha|}$.) Although both terms $\int_{\epsilon}^{\infty} \phi(v+s)$

$s)v^{-|\alpha|-1}dv$ and $\epsilon^{-|\alpha|}$ diverge as $\epsilon \rightarrow 0$, the limit of J_ϵ may exist. Moreover, since $\epsilon^{-|\alpha|} = |\alpha| \int_\epsilon^\infty v^{-|\alpha|-1}dv$, we get that

$$\lim_{\epsilon \rightarrow 0} J_\epsilon = \frac{|\alpha|}{\Gamma(1-|\alpha|)} \int_0^\infty (\phi(s) - \phi(v+s))v^{-|\alpha|-1}dv = (\mathbf{D}_-^\alpha \phi)(s). \quad (6.1.34)$$

The limit of J_ϵ is called the *finite part* of the divergent integral $I_-^\alpha \phi$ in the sense of Hadamard. The notation is $\lim_{\epsilon \rightarrow 0} J_\epsilon = \text{p.f. } I_-^\alpha \phi$, where p.f. stands for *partie finie*. By (6.1.34), we have

$$\mathbf{D}_-^\alpha \phi = \text{p.f. } I_-^\alpha \phi$$

(for sufficiently good function ϕ). A similar argument works for $\mathbf{D}_+^\alpha$.

The following theorem is proved in Samko et al. [878], p. 125. We will use it in the sequel with $\alpha \in (0, 1/2)$ and $p = 2$. The result is the real line counterpart of Proposition 6.1.8, (i).

Theorem 6.1.20 *If $f = I_\pm^\alpha \phi$ with $\phi \in L^p(\mathbb{R})$, $\alpha \in (0, 1)$ and $1 \leq p < 1/\alpha$, then $\mathbf{D}_\pm^\alpha f = \phi$; that is, $\mathbf{D}_\pm^\alpha I_\pm^\alpha \phi = \phi$.*

Thus to prove that $\mathbf{D}_\pm^\alpha f = \phi$, it is sufficient to show that $f = I_\pm^\alpha \phi$, as long as $\phi \in L^p(\mathbb{R})$, $1 \leq p < 1/\alpha$.

Remark 6.1.21 Because of Theorem 6.1.20 we will also use in the sequel the notation

$$I_\pm^\alpha = \mathbf{D}_\pm^{-\alpha} \quad \text{for } \alpha \in (-1, 0).$$

Thus, in the context of fractional integrals on the real line, $I_\pm^\alpha$ with $\alpha \in (-1, 0)$ will refer to the Marchaud and not the Liouville fractional derivative, unless mentioned otherwise.

Example 6.1.22 (*Marchaud fractional derivatives of indicator functions*) Let us show that, for $\alpha \in (0, 1)$ and $a < b$,

$$\Gamma(1-\alpha)(\mathbf{D}_\pm^\alpha 1_{[a,b)})(u) = (b-u)_\mp^{-\alpha} - (a-u)_\mp^{-\alpha}, \quad u \in \mathbb{R}, \quad (6.1.35)$$

which is similar to what one obtains with $\mathcal{D}_\pm^\alpha 1_{[a,b)}$ (compare (6.1.35) with (6.1.25)). By Theorem 6.1.20, it is enough to show that $I_\pm^\alpha((b-u)_\mp^{-\alpha} - (a-u)_\mp^{-\alpha}) = 1_{[a,b)}/\Gamma(1-\alpha)$. For I_-^α , we have

$$I = (I_-^\alpha((b-u)_+^{-\alpha} - (a-u)_+^{-\alpha}))(s) = \frac{1}{\Gamma(\alpha)} \int_{\mathbb{R}} ((b-u)_+^{-\alpha} - (a-u)_+^{-\alpha})(u-s)_+^{\alpha-1} du.$$

If $s \geq b$, then clearly $I = 0 = 1_{[a,b)}(s)/\Gamma(1-\alpha)$. If $a \leq s < b$, then

$$I = \frac{1}{\Gamma(\alpha)} \int_s^b (b-u)^{-\alpha}(u-s)^{\alpha-1} du = \frac{B(\alpha, 1-\alpha)}{\Gamma(\alpha)} = \frac{1_{[a,b)}(s)}{\Gamma(1-\alpha)}.$$

If $s < a$, we similarly get that $I = 0 = 1_{[a,b)}(s)/\Gamma(1-\alpha)$. Another way to prove (6.1.35) is by direct computations using Definition 6.1.18 (Exercise 6.1).

Remark 6.1.23 For $\alpha \geq 1$, the (left-sided) fractional derivative on the real line can be defined by extending (6.1.18), (6.1.28) or using the renormalization idea namely as

$$(\mathcal{D}_+^\alpha f)(u) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{du^n} \int_{-\infty}^u f(s)(u-s)^{n-1-\alpha} ds, \quad (6.1.36)$$

$$(\mathbf{D}_+^\alpha f)(u) = -\frac{1}{\Gamma(-\alpha)A_l(\alpha)} \int_0^\infty \frac{(\Delta_\xi^l f)(u)}{\xi^{1+\alpha}} d\xi, \quad (6.1.37)$$

where $n = [\alpha] + 1$ in (6.1.36) and (6.1.37) ($[\alpha]$ denotes the integer part of $\alpha > 0$), l in (6.1.37) is some integer such that $l > \alpha$, and

$$\begin{aligned} (\Delta_\xi^l f)(u) &= ((E - \tau_\xi)^l f)(u) = \sum_{k=0}^l (-1)^k \binom{l}{k} f(u - k\xi), \\ (Ef)(u) &= f(u), \quad (\tau_\xi^k f)(u) = f(u - k\xi), \quad u, \xi \in \mathbb{R}, \quad k \in \mathbb{N}, \\ A_l(\alpha) &= \sum_{k=1}^l (-1)^{k-1} \binom{l}{k} k^\alpha. \end{aligned}$$

The notation is similar to that used in time series analysis. E stands for the identity operator, τ_ξ for a shift to the left of length ξ (backshift operator), τ_ξ^k stands for the iterated shift and Δ_ξ is the difference operator $E - \tau_\xi$. The functions $\mathcal{D}_+^\alpha f$ and $\mathbf{D}_+^\alpha f$ are identical for good enough functions f (in particular, $\mathbf{D}_+^\alpha f$ does not depend on the choice of l , $l > \alpha$). For more details, see Samko et al. [878], pp. 114–120.

6.1.5 The Fourier Transform Perspective

It is helpful to use Fourier transforms to express fractional integrals and derivatives. They often provide a great simplification. Let $\widehat{\phi}(x) = \int_{\mathbb{R}} e^{ixu} \phi(u) du$ denote the $L^2(\mathbb{R})$ -Fourier transform of the function ϕ (see Appendix A.1.2). It is part of the folklore to write

$$\widehat{I_\pm^\alpha \phi}(x) = \widehat{\phi}(x) / (\mp ix)^\alpha, \quad (6.1.38)$$

where

$$(\mp ix)^\alpha = e^{\alpha \log |x| \mp i \operatorname{sign}(x) \frac{\alpha\pi}{2}} = |x|^\alpha e^{\mp i \operatorname{sign}(x) \frac{\alpha\pi}{2}}.$$

Observe that the semigroup property (6.1.26) can be verified heuristically if we use (6.1.38) to take the Fourier transforms of the left-hand side of (6.1.26) to obtain the Fourier transform of the right-hand side of (6.1.26). The idea behind the relation (6.1.38) is as follows. Consider, for example, the fractional integral I_+^α with $\alpha \in (0, 1/2)$. Then

$$(I_+^\alpha \phi)(s) = \frac{1}{\Gamma(\alpha)} \int_{\mathbb{R}} \phi(u)(s-u)_+^{\alpha-1} du \quad (6.1.39)$$

is well-defined for $\phi \in L^2(\mathbb{R})$ by the argument following Definition 6.1.14. Now observe that $I_+^\alpha \phi$ is the convolution of the functions $\phi(u)$ and $\psi_\alpha(u) = u_+^{\alpha-1} / \Gamma(\alpha)$. Since a convolution in the “time domain” becomes a product in the “spectral domain,” we would expect that

$$\widehat{I_+^\alpha \phi}(x) = \widehat{\phi}(x) \widehat{\psi}_\alpha(x).$$

(Observe, however, that the function ψ_a is neither in $L^1(\mathbb{R})$ nor in $L^2(\mathbb{R})$.) Taking the preceding relation nevertheless for granted, we obtain (6.1.38) by evaluating $\widehat{\psi}_a$ as follows:

$$\begin{aligned}\widehat{\psi}_a(x) &= \frac{1}{\Gamma(\alpha)} \int_{\mathbb{R}} e^{ixu} u_+^{\alpha-1} du \\ &= \frac{|x|^{-\alpha}}{\Gamma(\alpha)} \left(\int_{\mathbb{R}} e^{iv} v_+^{\alpha-1} dv 1_{\{x>0\}} + \int_{\mathbb{R}} e^{-iv} v_+^{\alpha-1} dv 1_{\{x<0\}} \right) = |x|^{-\alpha} e^{i \operatorname{sign}(x) \frac{\alpha\pi}{2}},\end{aligned}$$

since $\int_{\mathbb{R}} e^{\pm iv} v_+^{\alpha-1} dv = \Gamma(\alpha) e^{\pm i\alpha\pi/2}$ for $0 < \alpha < 1$ by (2.6.6). These arguments can be made rigorous by using approximations. We have the following result, which we state without proof. See, for example, Proposition 2.3 in Pipiras and Taqqu [815].

Proposition 6.1.24 *Let $\phi \in L^2(\mathbb{R})$. If $0 < \alpha < 1/2$, suppose that the function ϕ is such that*

$$\int_{\mathbb{R}} |\widehat{\phi}(x)|^2 |x|^{-2\alpha} dx < \infty. \quad (6.1.40)$$

If $-1/2 < \alpha < 0$, suppose that there is $f \in L^2(\mathbb{R})$ such that $\phi = I_{\pm}^{-\alpha} f$. Then, for $-1/2 < \alpha < 1/2$, we have that

$$\widehat{I_{\pm}^{-\alpha} \phi}(x) = \widehat{\phi}(x) / (\mp ix)^{\alpha}.$$

The conditions of Proposition 6.1.24 are practical. Suppose first $\alpha \in (0, 1/2)$. Since we are considering $L^2(\mathbb{R})$ -Fourier transforms, we want both the function ϕ and the result of the integration $\widehat{I_{\pm}^{-\alpha} \phi}(x) = \widehat{\phi}(x) / (\mp ix)^{\alpha}$ to be in $L^2(\mathbb{R})$, which gives the condition (6.1.40). The conditions for the case $\alpha \in (-1/2, 0)$ yield the result by allowing the conditions for $\alpha \in (0, 1/2)$ to be satisfied.

Example 6.1.25 (*Fourier transform of fractional integrals of indicator functions*) Let $\alpha \in (-1/2, 1/2)$ and $a, b \in \mathbb{R}$. We will show that the function $\phi(u) = 1_{[a,b]}(u)$, $u \in \mathbb{R}$, satisfies the conditions of Proposition 6.1.24. Indeed, if $\alpha \in (0, 1/2)$, then

$$\int_{\mathbb{R}} |\widehat{1}_{[a,b]}(x)|^2 |x|^{-2\alpha} dx = \int_{\mathbb{R}} \left| \frac{e^{ibx} - e^{iax}}{ix} \right|^2 |x|^{-2\alpha} dx < \infty.$$

If $\alpha \in (-1/2, 0)$, one can use (6.1.35) and verify by direct calculations (Exercise 6.2) that

$$1_{[a,b]} = I_{\pm}^{-\alpha} f_{a,b}, \quad (6.1.41)$$

where $f_{a,b} = I_{\pm}^{\alpha} 1_{[a,b]} = \mathbf{D}_{\pm}^{-\alpha} 1_{[a,b]}$ belongs to $L^2(\mathbb{R})$. Hence, Proposition 6.1.24 implies that

$$\widehat{I_{\pm}^{-\alpha} 1}_{[a,b]}(x) = \widehat{1}_{[a,b]}(x) (\mp ix)^{-\alpha} = \frac{e^{ibx} - e^{iax}}{ix} (\mp ix)^{-\alpha}. \quad (6.1.42)$$

Remark 6.1.26 Another interesting approach to fractional integro-differentiation is via fractional differences. For $\alpha > 0$, define²

$$f_{\pm}^{\alpha}(u) = \lim_{h \rightarrow 0} \frac{(\Delta_{\pm h}^{\alpha} f)(u)}{h^{\alpha}}, \quad (6.1.43)$$

where

$$(\Delta_h^{\alpha} f)(u) = ((E - \tau_h)^{\alpha} f)(u) = \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} f(u - kh),$$

$\binom{\alpha}{k}$ are the binomial coefficients, $(Ef)(u) = f(u)$ and $(\tau_h^k f)(u) = f(u - kh)$ for all $u, h \in \mathbb{R}$ and $k \geq 1$. The expressions in (6.1.43) are called *Grünwald–Letnikov fractional derivatives* of order α of a function f . For more information, see Section 20 in Samko et al. [878].

6.2 Representations of Fractional Brownian Motion

A number of integral representations of fractional Brownian motion (FBM) B_H were derived in previous chapters: the time domain representation in Proposition 2.6.5, the spectral domain representation in Proposition 2.6.11, and the finite interval representation in Proposition 4.2.5 (the latter when $H > 1/2$). We rewrite here these representations in terms of fractional integrals (derivatives), and also extend the finite interval representation to the case $H < 1/2$. We begin with and focus on the finite interval representation, which plays a central role in integration with respect to FBM.

Notation: In the context of fractional integration it is convenient to use another parametrization of FBM. Let

$$\kappa = H - \frac{1}{2}. \quad (6.2.1)$$

As in (2.8.4), κ is often denoted d , but d can be confusing in the integration context. Now $H \in (0, 1)$ corresponds to $\kappa \in (-1/2, 1/2)$. We will also denote the FBM B_H in terms of the parameter κ as B^{κ} :

$$B_H, \quad H \in (0, 1) \quad \Leftrightarrow \quad B^{\kappa}, \quad \kappa \in (-1/2, 1/2). \quad (6.2.2)$$

The usual standard Brownian motion is denoted B^0 .

While the case $\kappa = 0$ is always included below, our focus is on $\kappa \in (-1/2, 1/2) \setminus \{0\}$. In particular, in the proofs below, we will not deal with the case $\kappa = 0$, for which most of the results are trivial.

6.2.1 Representation of FBM on an Interval

We provide below the finite interval representation of FBM in terms of fractional integrals (derivatives) on an interval (these fractional integrals are defined in Section 6.1.1). The representation will involve the kernel function

² The notation $f_{\pm}^{\alpha}$ is not a typo!

$$f_t(s) = s^{-\kappa} \left(I_{a-}^{\kappa} u^{\kappa} 1_{[0,t)}(u) \right)(s), \quad (6.2.3)$$

where $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. By the definitions (6.1.3) and (6.1.12) of fractional integrals and derivatives, one has

$$f_t(s) = \frac{1}{\Gamma(\kappa)} s^{-\kappa} \int_0^t u^{\kappa} (u-s)_{+}^{\kappa-1} du, \quad \text{if } \kappa \in (0, 1/2), \quad (6.2.4)$$

$$f_t(s) = -\frac{1}{\Gamma(\kappa+1)} s^{-\kappa} \frac{d}{ds} \int_0^t u^{\kappa} (u-s)_{+}^{\kappa} du, \quad \text{if } \kappa \in (-1/2, 0). \quad (6.2.5)$$

We will first provide several alternative representations of the kernel function $f_t(s)$ which will be useful in the sequel.

Proposition 6.2.1 *Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. Let also $f_t(s)$ be defined in (6.2.3). Then,*

$$-\Gamma(\kappa+1)f_t(s) = s^{-\kappa} \frac{d}{ds} \int_0^t u^{\kappa} (u-s)_{+}^{\kappa} du \quad (6.2.6)$$

$$= (2\kappa+1)s^{-\kappa-1} \int_s^t u^{\kappa} (u-s)_{+}^{\kappa} du - \left(\frac{t}{s}\right)^{\kappa+1} (t-s)_{+}^{\kappa} \quad (6.2.7)$$

$$= \kappa s^{-\kappa} \int_s^t u^{\kappa-1} (u-s)_{+}^{\kappa} du - \left(\frac{t}{s}\right)^{\kappa} (t-s)_{+}^{\kappa} \quad (6.2.8)$$

$$= s^{\kappa} F_i\left(\frac{t}{s}\right) 1_{[0,t)}(s), \quad i = 1, 2, \quad (6.2.9)$$

where, for $z > 1$,

$$F_1(z) = (2\kappa+1) \int_1^z w^{\kappa} (w-1)^{\kappa} dw - z^{\kappa+1} (z-1)^{\kappa}, \quad (6.2.10)$$

$$F_2(z) = \kappa \int_1^z w^{\kappa-1} (w-1)^{\kappa} dw - z^{\kappa} (z-1)^{\kappa}. \quad (6.2.11)$$

Remark 6.2.2 When $s \geq t$, all sides of (6.2.6)–(6.2.8) are equal to 0, and therefore it is enough to define $F_1(z)$ and $F_2(z)$ for $z > 1$.

Remark 6.2.3 The relation (6.2.8) can informally be derived as

$$\begin{aligned} \frac{d}{ds} \int_0^t u^{\kappa} (u-s)_{+}^{\kappa} du &= \frac{d}{ds} \int_s^t u^{\kappa} (u-s)_{+}^{\kappa} du = \frac{d}{ds} \int_0^t (v+s)^{\kappa} 1_{\{v < t-s\}} v^{\kappa} dv \\ &= \kappa \int_0^t (v+s)^{\kappa-1} 1_{\{v < t-s\}} v^{\kappa} dv + \int_0^t (v+s)^{\kappa} (-\delta_{\{t-s\}}(v)) v^{\kappa} dv \\ &= \kappa \int_s^t u^{\kappa-1} (u-s)_{+}^{\kappa} du - t^{\kappa} (t-s)_{+}^{\kappa}, \end{aligned}$$

where $\delta_{\{a\}}(v)$ denotes a point mass at $v = a$.

Proof It is enough to consider the case $0 < s < t$. When $\kappa \in (-1/2, 0)$, the relation (6.2.6) is given above. When $\kappa \in (0, 1/2)$, by differentiating inside the integral, we expect that

$$s^{-\kappa} \frac{d}{ds} \int_0^t u^\kappa (u-s)_+^\kappa du = -\kappa s^{-\kappa} \int_0^t u^\kappa (u-s)_+^{\kappa-1} du.$$

This could be proved by the definition of derivative. For example, for $h > 0$, consider

$$\begin{aligned} \int_0^t u^\kappa (u-s-h)_+^\kappa du - \int_0^t u^\kappa (u-s)_+^\kappa du &= \int_{s+h}^t u^\kappa \left((u-s-h)^\kappa - (u-s)^\kappa \right) du \\ &\quad - \int_s^{s+h} u^\kappa (u-s)^\kappa du =: T_1(s, h) + T_2(s, h). \end{aligned}$$

By the mean value theorem,

$$\frac{T_1(s, h)}{h} = \kappa \int_{s+h}^t u^\kappa (u-s^*(h))^{\kappa-1} du = \kappa \int_{s+h-s^*(h)}^{t-s^*(h)} (u+s^*(h))^\kappa u^{\kappa-1} du,$$

where $s < s^*(h) < s+h$. As $h \rightarrow 0$, $T_1(s, h)/h \rightarrow \kappa \int_0^{t-s} (u+s)^\kappa u^{\kappa-1} du = \kappa \int_0^t u^\kappa (u-s)_+^{\kappa-1} du$ by the dominated convergence theorem. On the other hand, by making the change of variables $u-s = hv$, $T_2(s, h) = -h^{\kappa+1} \int_0^1 (s+hv)^\kappa v^\kappa dv \sim C(s)h^{\kappa+1}$ and hence $T_2(s, h)/h \rightarrow 0$, as $h \rightarrow 0$.

Note also that the expression (6.2.9) with $i = 1$ is the expression (6.2.7), and that (6.2.9) with $i = 2$ is the expression (6.2.8). It is therefore enough to show that $F_1(z) = F_2(z)$, for $z > 1$, and that the relation (6.2.9) holds with $i = 1$.

The relation (6.2.9) with $i = 1$ holds since, after the change of variables $u = sw$,

$$\begin{aligned} \frac{d}{ds} \int_0^t u^\kappa (u-s)_+^\kappa du &= \frac{d}{ds} \int_s^t u^\kappa (u-s)^\kappa du = \frac{d}{ds} s^{2\kappa+1} \int_1^{t/s} w^\kappa (w-1)_+^\kappa dw \\ &= (2\kappa+1)s^{2\kappa} \int_1^{t/s} w^\kappa (w-1)_+^\kappa dw - s^{2\kappa+1} \frac{t}{s^2} \left(\frac{t}{s}\right)^\kappa \left(\frac{t}{s}-1\right)^\kappa = s^{2\kappa} F_1\left(\frac{t}{s}\right). \end{aligned}$$

Finally, for $F_1(z) = F_2(z)$, $z > 1$, note first that

$$\begin{aligned} \frac{d}{dz} F_1(z) &= (2\kappa+1)z^\kappa (z-1)^\kappa - (\kappa+1)z^\kappa (z-1)^\kappa - \kappa z^{\kappa+1} (z-1)^{\kappa-1} \\ &= \kappa z^\kappa (z-1)^\kappa - \kappa z^{\kappa+1} (z-1)^{\kappa-1} = -\kappa z^\kappa (z-1)^\kappa, \\ \frac{d}{dz} F_2(z) &= \kappa z^{\kappa-1} (z-1)^\kappa - \kappa z^{\kappa-1} (z-1)^\kappa - \kappa z^\kappa (z-1)^{\kappa-1} = -\kappa z^\kappa (z-1)^\kappa, \end{aligned}$$

so that $\frac{d}{dz} F_1(z) = \frac{d}{dz} F_2(z)$. To conclude that $F_1(z) = F_2(z)$, it is enough to show that $\lim_{z \downarrow 1} (F_1(z) - F_2(z)) = 0$. When $\kappa \in (0, 1/2)$, this follows from $F_1(1+) = F_2(1+) = 0$. For $\kappa \in (-1/2, 0)$, this follows from noting that

$$F_1(z) - F_2(z) = (2\kappa+1) \int_1^z w^\kappa (w-1)^\kappa dw - \kappa \int_1^z w^{\kappa-1} (w-1)^\kappa dw - z^\kappa (z-1)^{\kappa+1},$$

since $(z^{\kappa+1} - z^\kappa)(z-1)^\kappa = z^\kappa (z-1)^{\kappa+1}$, which tends to 0, as $z \downarrow 1$. $\square$

The following three results are immediate corollaries of Proposition 6.2.1. The first corollary shows that the derivative of the kernel function $f_t(s)$ with respect to t has a particularly simple form. This result is quite useful (see, for example, the proof of Proposition 6.2.8 in the case $\kappa \in (-1/2, 0)$).

Corollary 6.2.4 Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. Let also $f_t(s)$ be defined in (6.2.3). Then,

$$\frac{\partial}{\partial t} f_t(s) = \frac{s^{-\kappa} t^{\kappa} (t-s)^{\kappa-1}}{\Gamma(\kappa)}. \quad (6.2.12)$$

Proof By using (6.2.8),

$$\begin{aligned} -\Gamma(\kappa+1) \frac{\partial}{\partial t} f_t(s) &= \kappa s^{-\kappa} t^{\kappa-1} (t-s)^{\kappa} - \kappa \frac{t^{\kappa-1}}{s^{\kappa}} (t-s)^{\kappa} - \kappa \left(\frac{t}{s}\right)^{\kappa} (t-s)^{\kappa-1} \\ &= -\kappa s^{-\kappa} t^{\kappa} (t-s)^{\kappa-1}, \end{aligned}$$

which yields the result. $\square$

The second corollary provides the behavior of the kernel function around the points $s = 0$ and $t = 0$.

Corollary 6.2.5 Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. Let also $f_t(s)$ be defined in (6.2.3). Then, as $s \downarrow 0$,

$$f_t(s) \sim \begin{cases} \frac{t^{2\kappa}}{2\Gamma(\kappa+1)} s^{-\kappa}, & \text{if } \kappa \in (0, 1/2), \\ \frac{\Gamma(1-2\kappa)}{2\Gamma(1-\kappa)} s^{\kappa}, & \text{if } \kappa \in (-1/2, 0) \end{cases} \quad (6.2.13)$$

and, as $s \uparrow t$,

$$f_t(s) \sim \frac{(t-s)^{\kappa}}{\Gamma(\kappa+1)}. \quad (6.2.14)$$

Proof The relation (6.2.13) follows from (6.2.9) after observing that as $z \rightarrow \infty$, $F_2(z) \sim \kappa \int_1^z w^{2\kappa-1} dw - z^{2\kappa} \sim -z^{2\kappa}/2$ when $\kappa \in (0, 1/2)$, and that $F_2(z) \sim \kappa \int_1^{\infty} w^{\kappa-1} (w-1)^{\kappa} dw = \kappa \int_0^{\infty} z^{\kappa} (z+1)^{\kappa-1} dz = \kappa B(\kappa+1, -2\kappa) = \kappa \frac{\Gamma(\kappa+1)\Gamma(-2\kappa)}{\Gamma(1-\kappa)} = -\frac{\Gamma(\kappa+1)\Gamma(1-2\kappa)}{2\Gamma(1-\kappa)}$ when $\kappa \in (-1/2, 0)$. The relation (6.2.14) follows from (6.2.9) after observing that as $z \downarrow 1$, $F_2(z) \sim -(z-1)^{\kappa}$. $\square$

The third corollary provides yet another, useful representation of the kernel function.

Corollary 6.2.6 Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. Let also $f_t(s)$ be defined in (6.2.3). Then,

$$-\Gamma(\kappa+1) f_t(s) = \kappa s^{-\kappa} \int_s^t (s^{\kappa} - u^{\kappa})(u-s)^{\kappa-1}_+ du - (t-s)^{\kappa-1}_+. \quad (6.2.15)$$

Proof The relation (6.2.15) can be proved by rewriting the right-hand side of (6.2.8) as follows:

$$\begin{aligned} &\kappa s^{-\kappa} \int_s^t u^{\kappa-1} (u-s)^{\kappa}_+ du - \left(\frac{t}{s}\right)^{\kappa} (t-s)^{\kappa}_+ \\ &= \kappa s^{-\kappa} \int_s^t u^{\kappa-1} \left((u-s)^{\kappa}_+ - (t-s)^{\kappa}_+\right) du + \kappa s^{-\kappa} \int_s^t u^{\kappa-1} du (t-s)^{\kappa}_+ - s^{-\kappa} t^{\kappa} (t-s)^{\kappa}_+ \end{aligned}$$

$$\begin{aligned}
&= -\kappa^2 s^{-\kappa} \int_s^t u^{\kappa-1} \left(\int_u^t (z-s)_+^{\kappa-1} dz \right) du + s^{-\kappa} (t^\kappa - s^\kappa) (t-s)_+^\kappa - s^{-\kappa} t^\kappa (t-s)_+^\kappa \\
&= -\kappa^2 s^{-\kappa} \int_s^t (z-s)_+^{\kappa-1} \left(\int_s^z u^{\kappa-1} du \right) dz - (t-s)_+^\kappa \\
&= -\kappa s^{-\kappa} \int_s^t (z^\kappa - s^\kappa) (z-s)_+^{\kappa-1} dz - (t-s)_+^\kappa,
\end{aligned}$$

where we changed the order of integration above. $\square$

The next alternative representation involves the celebrated Gauss hypergeometric function ${}_2F_1(a, b, c, z)$. Recall that this function is defined for any $a, b, c, z \in \mathbb{C}$ satisfying $c \neq 0, -1, \dots$, and $|\arg(1-z)| < \pi$ (that is, z is not on the section of the real axis starting at $+1$ and stretching to $+\infty$; any such z would satisfy $\arg(1-z) = -\pi$). When $|z| \leq 1$, the function is given by the series

$${}_2F_1(a, b, c, z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{z^k}{k!}, \quad (6.2.16)$$

where $(a)_0 = 1$ and $(a)_k = \Gamma(a+k)/\Gamma(a) = a(a+1)\dots(a+k-1)$ is the Pochhammer symbol. For other z 's (satisfying $|\arg(1-z)| < \pi$), the hypergeometric function is extended by analytic continuation. When $0 < \Re b < \Re c$, it is known that

$${}_2F_1(a, b, c, z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 v^{b-1} (1-v)^{c-b-1} (1-zv)^{-a} dv, \quad (6.2.17)$$

for all z satisfying $|\arg(1-z)| < \pi$. It is also known that, for fixed z , ${}_2F_1(a, b, c, z)$ is analytic in its parameters a, b and c . This and more information on the Gauss hypergeometric function can be found, for example, in Lebedev [601], Section 9.

Proposition 6.2.7 *Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and $s, t \in [0, a]$. Let also $f_t(s)$ be defined in (6.2.3). Then,*

$$\Gamma(\kappa+1) f_t(s) = (t-s)_+^\kappa {}_2F_1\left(-\kappa, \kappa, \kappa+1, 1-\frac{t}{s}\right). \quad (6.2.18)$$

Proof When $\kappa \in (0, 1/2)$ and $0 \leq s < t \leq a$, the representation follows from

$$\begin{aligned}
\Gamma(\kappa+1) f_t(s) &= \kappa s^{-\kappa} \int_s^t u^\kappa (u-s)^{\kappa-1} du = \kappa s^{-\kappa} \int_0^{t-s} (v+s)^\kappa v^{\kappa-1} dv \\
&= \kappa s^{-\kappa} (t-s)^\kappa \int_0^1 ((t-s)w+s)^\kappa w^{\kappa-1} dw \\
&= \kappa (t-s)^\kappa \int_0^1 w^{\kappa-1} \left(1 - \left(1 - \frac{t}{s}\right)w\right)^\kappa dw \\
&= (t-s)^\kappa \frac{\Gamma(\kappa+1)}{\Gamma(\kappa)\Gamma((\kappa+1)-\kappa)} \int_0^1 w^{\kappa-1} (1-w)^{(\kappa+1)-\kappa-1} \left(1 - \left(1 - \frac{t}{s}\right)w\right)^\kappa dw \\
&= (t-s)^\kappa {}_2F_1\left(-\kappa, \kappa, \kappa+1, 1-\frac{t}{s}\right),
\end{aligned}$$

where we used the changes of variables $v = su$ and $v = (t - s)w$, and the integral representation (6.2.17).

When $\kappa \in (-1/2, 0)$, the relation (6.2.18) can be proved by arguing that the left- and right-hand sides are analytic functions for $|\kappa| < 1/2$, $\kappa \in \mathbb{C}$. Then, since the two sides coincide for real $\kappa \in (0, 1/2)$, they also coincide for real $\kappa \in (-1/2, 0)$. For the right-hand side, the analyticity follows from the analytic properties of the hypergeometric function mentioned above. For the left-hand side, suppose it to be defined, for example, by the representation (6.2.7), which is now considered for $|\kappa| < 1/2$, $\kappa \in \mathbb{C}$. All power functions a^κ , $a \neq 0$, entering (6.2.7) are analytic. The integral $f(\kappa) = \int_s^t u^\kappa (u - s)^\kappa du$ can be shown to be analytic for $|\kappa| < 1/2$ directly through the definition by using the arguments as those in Pipiras and Taqqu [817], Appendix A. For example, as in that appendix, one can show that

$$\frac{d}{d\kappa} f(\kappa) = \int_s^t \left(\log u u^\kappa (u - s)^\kappa + \log(u - s) u^\kappa (u - s)^\kappa \right) du. \quad \square$$

Most mathematical software, for example, MATLAB, performs calculations of Gauss hypergeometric function. Figure 6.2 plots the kernel function $f_t(s)$ for $t = 1$ and several values of κ .

We now provide the finite interval representation of FBM in terms of fractional integrals (derivatives) on an interval. It involves the function $f_t(s)$ in (6.2.3). When $\kappa \in (0, 1/2)$, the representation corresponds to that in Proposition 4.2.5, but we will provide an alternative and more direct proof.

Proposition 6.2.8 *Let $a > 0$, $\kappa \in (-1/2, 1/2)$ and B^κ be a standard FBM. Then,*

$$\{B^\kappa(t)\}_{t \in [0, a]} \stackrel{d}{=} \left\{ \sigma_\kappa \int_0^a s^{-\kappa} \left(I_{a-}^\kappa u^\kappa 1_{[0, t)}(u) \right) (s) dB^0(s) \right\}_{t \in [0, a]} \quad (6.2.19)$$

in the sense of finite-dimensional distributions, where

$$\sigma_\kappa^2 = \frac{\pi \kappa (2\kappa + 1)}{\Gamma(1 - 2\kappa) \sin(\kappa \pi)} = \frac{\Gamma(\kappa)^2 \kappa (2\kappa + 1)}{B(\kappa, 1 - 2\kappa)}. \quad (6.2.20)$$

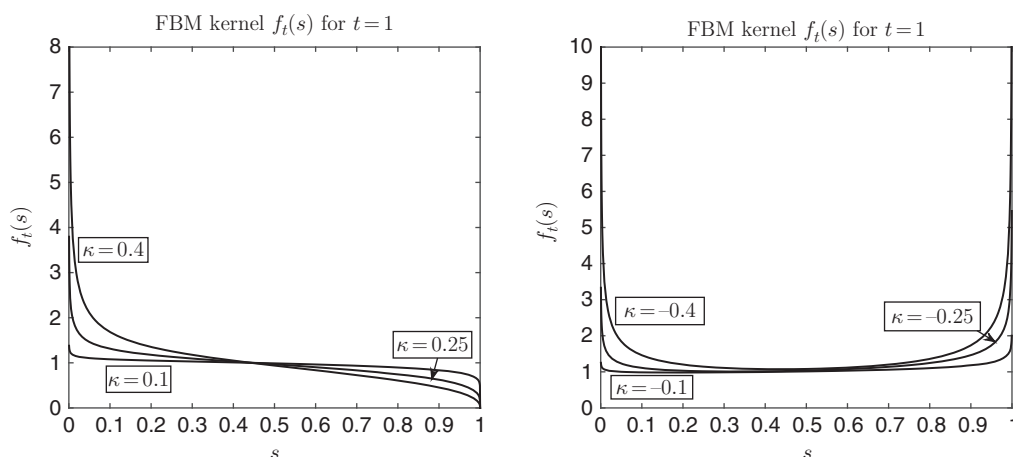


Figure 6.2 The kernel function $f_t(s)$ in (6.2.3) for $t = 1$ and several values of κ .

Remark 6.2.9 Note that the endpoint a of the interval $[0, a]$ plays little role in the representation (6.2.19). The kernel function $f_t(s)$ in (6.2.3) is supported on the interval $[0, t]$. It could be expressed alternatively as

$$f_t(s) = s^{-\kappa} \left(I_-^\kappa u^\kappa 1_{[0,t)}(u) \right)(s),$$

where I_-^κ is a fractional integral or derivative in (6.1.17) or (6.1.30). The processes on the two sides of (6.2.19) can then be indexed by $t \in [0, \infty)$, instead of $t \in [0, a]$, and the integral $\int_0^a$ replaced by $\int_0^\infty$. (Another possibility is to use I_{t-}^κ and the integral $\int_0^t$.) In any case, the integration is effectively over the interval $[0, t]$. It is for this reason that (6.2.19) is called a representation on an interval.

Proof The cases $\kappa \in (0, 1/2)$ and $\kappa \in (-1/2, 0)$ are treated separately.

The case $\kappa \in (0, 1/2)$: Since the covariance function of a standard FBM B^κ is

$$\Gamma^\kappa(u, v) = \mathbb{E} B^\kappa(u) B^\kappa(v) = \frac{1}{2} \left(|u|^{2\kappa+1} + |v|^{2\kappa+1} - |u-v|^{2\kappa+1} \right), \quad (6.2.21)$$

with $u, v \in \mathbb{R}$, we get that $d^2 \Gamma^\kappa(u, v) = \kappa(2\kappa+1)|u-v|^{2\kappa-1} du dv$. This implies that

$$\Gamma^\kappa(t_1, t_2) = \kappa(2\kappa+1) \int_{\mathbb{R}} \int_{\mathbb{R}} 1_{[0,t_1)}(u) 1_{[0,t_2)}(v) |u-v|^{2\kappa-1} du dv. \quad (6.2.22)$$

We will argue next³ that, for $u, v \in [0, a]$,

$$|u-v|^{2\kappa-1} = \frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_0^a s^{-2\kappa} (v-s)^{\kappa-1} (u-s)^{\kappa-1} ds, \quad (6.2.23)$$

where B is the beta function defined in (2.2.12). Suppose without loss of generality that $v > u$. Then, the right-hand side of (6.2.23) becomes

$$\frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_0^u s^{-2\kappa} (v-s)^{\kappa-1} (u-s)^{\kappa-1} ds = \frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_0^u \left(\frac{v}{s} - 1 \right)^{\kappa-1} \left(\frac{u}{s} - 1 \right)^{\kappa-1} \frac{ds}{s^2}.$$

After the change of variables $1/s = w$, this becomes

$$\frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_{1/u}^\infty (vw-1)^{\kappa-1} (uw-1)^{\kappa-1} dw.$$

Further changes of variables $uw-1 = y$ and then $y = \frac{(v/u-1)}{(v/u)}z = \frac{v-u}{v}z$ lead to

$$\begin{aligned} & \frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_0^\infty \left(\frac{v}{u}y + \left(\frac{v}{u} - 1 \right) \right)^{\kappa-1} y^{\kappa-1} \frac{dy}{u} \\ &= \frac{(uv)^\kappa}{B(\kappa, 1-2\kappa)} \int_0^\infty \left(\frac{v}{u} - 1 \right)^{\kappa-1} (z+1)^{\kappa-1} \left(\frac{v-u}{v} \right)^{\kappa-1} z^{\kappa-1} \frac{v-u}{v} \frac{dz}{u} \\ &= \frac{(v-u)^{2\kappa-1}}{B(\kappa, 1-2\kappa)} \int_0^\infty (z+1)^{\kappa-1} z^{\kappa-1} dz = (v-u)^{2\kappa-1}, \end{aligned}$$

³ See also the proof of Lemma 4.2.7.

since $\int_0^\infty (z+1)^{\kappa-1} z^{\kappa-1} dz = B(\kappa, 1-2\kappa)$ by using (2.2.12). This establishes (6.2.23). (One can also prove (6.2.23) by making the single change of variables $s = (uv)z/(\min(u, v)z + |u - v|)$.)

By changing the order of integration in (6.2.22) with $|u - v|^{2\kappa-1}$ substituted by (6.2.23) and using Definition 6.1.1, we obtain that, for $t_1, t_2 \in [0, a]$,

$$\Gamma^\kappa(t_1, t_2) = \frac{\Gamma(\kappa)^2 \kappa (2\kappa + 1)}{B(\kappa, 1 - 2\kappa)} \int_0^a \left(s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, t_1)}(u))(s) \right) \left(s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, t_2)}(u))(s) \right) ds.$$

This establishes the proposition in the case $\kappa \in (0, 1/2)$.

The case $\kappa \in (-1/2, 0)$: Let $f_t(s)$ be the kernel function defined in (6.2.3). We want to show that, for $0 \leq t_1 < t_2 \leq a$,

$$\int_0^a f_{t_1}(s) f_{t_2}(s) ds = \frac{1}{2\sigma_\kappa^2} \left(t_1^{2\kappa+1} + t_2^{2\kappa+1} - (t_2 - t_1)^{2\kappa+1} \right).$$

It is enough to prove that

$$\frac{\partial}{\partial t_2} \int_0^a f_{t_1}(s) f_{t_2}(s) ds = \frac{2\kappa + 1}{2\sigma_\kappa^2} \left(t_2^{2\kappa} - (t_2 - t_1)^{2\kappa} \right), \quad t_1 < t_2, \quad (6.2.24)$$

and when $t_2 = t_1$,

$$\int_0^a f_{t_1}(s) f_{t_1}(s) ds = \frac{t_1^{2\kappa}}{\sigma_\kappa^2}. \quad (6.2.25)$$

The relation (6.2.25) is left as an exercise (Exercise 6.3). By taking the differentiation inside the integral, the relation (6.2.24) would follow from

$$\int_0^a f_{t_1}(s) \frac{\partial}{\partial t_2} f_{t_2}(s) ds = \frac{2\kappa + 1}{2\sigma_\kappa^2} \left(t_2^{2\kappa} - (t_2 - t_1)^{2\kappa} \right). \quad (6.2.26)$$

By using (6.2.9) and (6.2.12), this is equivalent to

$$\int_0^{t_1} \frac{s^\kappa F_2(\frac{t_1}{s})}{-\Gamma(\kappa + 1)} \frac{s^{-\kappa} t_2^\kappa (t_2 - s)^{\kappa-1}}{\Gamma(\kappa)} ds = \frac{2\kappa + 1}{2\sigma_\kappa^2} \left(t_2^{2\kappa} - (t_2 - t_1)^{2\kappa} \right)$$

or, after the change of variables $s = t_1 w$,

$$\int_0^1 F_2\left(\frac{1}{w}\right) t_2^\kappa (t_2 - t_1 w)^{\kappa-1} t_1 dw = -\frac{(2\kappa + 1)\Gamma(\kappa + 1)\Gamma(\kappa)}{2\sigma_\kappa^2} \left(t_2^{2\kappa} - (t_2 - t_1)^{2\kappa} \right)$$

or, after setting $b = t_2/t_1$ and simplifying the constant by using (6.2.20),

$$\int_0^1 F_2\left(\frac{1}{w}\right) (b - w)^{\kappa-1} dw = \frac{B(\kappa, 1 - 2\kappa)}{2} b^{-\kappa} \left((b - 1)^{2\kappa} - b^{2\kappa} \right), \quad b > 1. \quad (6.2.27)$$

Denote the left-hand side of (6.2.27) by $I(b)$. We want to show that it is equal to the right-hand side. By using (6.2.6) and (6.2.9) with $t = 1$,

$$I(b) = \int_0^1 \frac{d}{dw} \left(\int_0^1 u^\kappa (u - w)_+^\kappa du \right) w^{-2\kappa} (b - w)^{\kappa-1} dw.$$

By integration by parts and then changing the order of integration, we get

$$\begin{aligned}
 I(b) &= - \int_0^1 \left(\int_0^1 u^\kappa (u-w)_+^\kappa du \right) \frac{d}{dw} \left(w^{-2\kappa} (b-w)^{\kappa-1} \right) dw \\
 &= 2\kappa \int_0^1 \left(\int_w^1 u^\kappa (u-w)^\kappa du \right) w^{-2\kappa-1} (b-w)^{\kappa-1} dw \\
 &\quad + (\kappa-1) \int_0^1 \left(\int_w^1 u^\kappa (u-w)^\kappa du \right) w^{-2\kappa} (b-w)^{\kappa-2} dw \\
 &= 2\kappa \int_0^1 du u^\kappa \int_0^u w^{-2\kappa-1} (u-w)^\kappa (b-w)^{\kappa-1} dw \\
 &\quad + (\kappa-1) \int_0^1 du u^\kappa \int_0^u w^{-2\kappa} (u-w)^\kappa (b-w)^{\kappa-2} dw.
 \end{aligned}$$

By making a series of changes of variables $w = uv$, $v = 1/z$, $z = x + 1$ and $x = \frac{(b/u)-1}{(b/u)}y = \frac{b-u}{b}y$ below, we further get that

$$\begin{aligned}
 I(b) &= 2\kappa \int_0^1 du u^{\kappa-1} \int_0^1 v^{-2\kappa-1} (1-v)^\kappa \left(\frac{b}{u} - v \right)^{\kappa-1} dv \\
 &\quad + (\kappa-1) \int_0^1 du u^{\kappa-1} \int_0^1 v^{-2\kappa} (1-v)^\kappa \left(\frac{b}{u} - v \right)^{\kappa-2} dv \\
 &= 2\kappa \int_0^1 du u^{\kappa-1} \int_1^\infty (z-1)^\kappa \left(\frac{b}{u}z - 1 \right)^{\kappa-1} dz \\
 &\quad + (\kappa-1) \int_0^1 du u^{\kappa-1} \int_1^\infty (z-1)^\kappa \left(\frac{b}{u}z - 1 \right)^{\kappa-2} dz \\
 &= 2\kappa \int_0^1 du u^{\kappa-1} \int_0^\infty x^\kappa \left(\frac{b}{u}x + \frac{b}{u} - 1 \right)^{\kappa-1} dx \\
 &\quad + (\kappa-1) \int_0^1 du u^{\kappa-1} \int_0^\infty x^\kappa \left(\frac{b}{u}x + \frac{b}{u} - 1 \right)^{\kappa-2} dx \\
 &= 2\kappa b^{-\kappa-1} \int_0^1 (b-u)^{2\kappa} du \int_0^\infty y^\kappa (y+1)^{\kappa-1} dy \\
 &\quad + (\kappa-1) b^{-\kappa-1} \int_0^1 u(b-u)^{2\kappa-1} du \int_0^\infty y^\kappa (y+1)^{\kappa-2} dy.
 \end{aligned}$$

In view of (2.2.12) and then by integration by parts, we obtain that

$$\begin{aligned}
 I(b) &= 2\kappa B(\kappa+1, -2\kappa) b^{-\kappa-1} \int_0^1 (b-u)^{2\kappa} du \\
 &\quad + (\kappa-1) B(\kappa+1, 1-2\kappa) b^{-\kappa-1} \int_0^1 u(b-u)^{2\kappa-1} du \\
 &= B(\kappa, 1-2\kappa) \kappa b^{-\kappa-1} \left(- \int_0^1 (b-u)^{2\kappa} du - \int_0^1 u(b-u)^{2\kappa-1} du \right) \\
 &= B(\kappa, 1-2\kappa) b^{-\kappa-1} \left(\frac{1}{2} (b-1)^{2\kappa} - \frac{2\kappa+1}{2} \int_0^1 (b-u)^{2\kappa} du \right)
 \end{aligned}$$

$$\begin{aligned}
&= B(\kappa, 1 - 2\kappa)b^{-\kappa-1} \left(\frac{1}{2}(b-1)^{2\kappa} - \frac{2\kappa+1}{2} \int_0^1 (b-u)^{2\kappa} du \right) \\
&= B(\kappa, 1 - 2\kappa)b^{-\kappa-1} \left(\frac{1}{2}(b-1)^{2\kappa} + \frac{1}{2}(b-1)^{2\kappa+1} - \frac{1}{2}b^{2\kappa+1} \right) \\
&= B(\kappa, 1 - 2\kappa)b^{-\kappa-1} \left(\frac{(b-1)^{2\kappa}b}{2} - \frac{b^{2\kappa}b}{2} \right) = \frac{B(\kappa, 1 - 2\kappa)}{2} b^{-\kappa} \left((b-1)^{2\kappa} - b^{2\kappa} \right),
\end{aligned}$$

which establishes (6.2.27). $\square$

6.2.2 Representations of FBM on the Real Line

In the next result, we rewrite the integral representations (2.6.1) and (2.6.9) of FBM using fractional integrals (derivatives) on the real line.

Proposition 6.2.10 *Let $\kappa \in (-1/2, 1/2)$. A standard FBM B^κ has the following representations in the sense of finite-dimensional distributions:*

(a) *Time domain representation:*

$$\{B^\kappa(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{1}{c_{1,\kappa}} \int_{\mathbb{R}} \left((t-s)_+^\kappa - (-s)_+^\kappa \right) dB^0(s) \right\}_{t \in \mathbb{R}} \quad (6.2.28)$$

$$\stackrel{d}{=} \left\{ \frac{\Gamma(\kappa+1)}{c_{1,\kappa}} \int_{\mathbb{R}} (I_-^\kappa 1_{[0,t)})(s) dB^0(s) \right\}_{t \in \mathbb{R}}, \quad (6.2.29)$$

where

$$c_{1,\kappa}^2 = \int_{\mathbb{R}} \left((1-u)_+^\kappa - (-u)_+^\kappa \right)^2 du = \frac{\kappa B(\kappa, 1-\kappa)}{2\kappa+1} \quad (6.2.30)$$

and $dB^0(s) = B^0(ds)$ is a Gaussian measure with the Lebesgue control measure $\mathbb{E}(B(ds))^2 = ds$.

(b) *Spectral domain representation:*

$$\{B^\kappa(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{\Gamma(\kappa+1)}{(2\pi)^{1/2} c_{1,\kappa}} \int_{\mathbb{R}} \frac{e^{ixt} - 1}{ix} |x|^{-\kappa} d\widehat{B}(x) \right\}_{t \in \mathbb{R}} \quad (6.2.31)$$

$$\stackrel{d}{=} \left\{ \frac{\Gamma(\kappa+1)}{(2\pi)^{1/2} c_{1,\kappa}} \int_{\mathbb{R}} \frac{e^{ixt} - 1}{ix} (ix)^{-\kappa} d\widehat{B}(x) \right\}_{t \in \mathbb{R}} \quad (6.2.32)$$

$$\stackrel{d}{=} \left\{ \frac{\Gamma(\kappa+1)}{(2\pi)^{1/2} c_{1,\kappa}} \int_{\mathbb{R}} (\widehat{I_-^\kappa 1_{[0,t)}})(x) d\widehat{B}(x) \right\}_{t \in \mathbb{R}}, \quad (6.2.33)$$

where $d\widehat{B}(x) = \widehat{B}(dx)$ is an Hermitian Gaussian random measure with the Lebesgue control measure $\mathbb{E}|\widehat{B}(dx)|^2 = dx$.

Proof The representation (6.2.28) of standard FBM is the one given in (2.6.1). The representation (6.2.29) follows from (6.1.24) when $\kappa \in (0, 1/2)$, and (6.1.35) when $\kappa \in (-1/2, 0)$. The constant $c_{1,\kappa}$ in (6.2.30) is given below (2.6.1), and the second equality in (6.2.30) follows from (2.6.12).

The spectral representation (6.2.31) is given in Proposition 2.6.11 and the normalizing factor follows from (2.6.11). Since $|(ix)^{-\kappa}|^2 = (|x|^{-\kappa})^2$, the process in (6.2.32) has the

same covariance as the one in (6.2.31). Finally, the representation (6.2.33) follows from relation (6.1.42). $\square$

6.3 Fractional Wiener Integrals and their Deterministic Integrands

6.3.1 The Gaussian Space Generated by Fractional Wiener Integrals

Fractional Wiener integrals on an interval $[0, a]$ are integrals of the form

$$\int_0^a f(u) dB^\kappa(u) =: \mathcal{I}_a^\kappa(f), \quad (6.3.1)$$

where f is a deterministic function and B^κ is FBM. In contrast to integration with respect to Brownian motion, FBM does not have independent increments when $\kappa \neq 0$.

We shall first discuss what kind of objects the integrals (6.3.1) are, and what purposes they serve. We focus on the case of an interval $[0, a]$ throughout, though the case of the real line will also be discussed briefly below.

As with most integrals, one can think informally of a fractional Wiener integral as

$$\mathcal{I}_a^\kappa(f) = \int_0^a f(u) dB^\kappa(u) \approx \sum_{k=0}^{K-1} f_k^K (B^\kappa(u_{k+1}^K) - B^\kappa(u_k^K)), \quad (6.3.2)$$

where $0 \leq u_0^K < u_1^K < \dots < u_K^K \leq a$ is a partition of $[0, a]$, and f_k^K is somehow related to $f(u)$, for example, $f_k^K = f(u_k^K)$ if f is sufficiently smooth. The fractional Wiener integral is expected to be defined by (6.3.2) in the limit of $K \rightarrow \infty$. It is natural to take here the limit in the $L^2(\Omega)$ -sense. Note also that the sum in (6.3.2) is a linear combination of the values of FBM on the interval $[0, a]$.

Motivated by these observations, it is natural to consider the following space of random variables.

Definition 6.3.1 The Gaussian space $\mathcal{H}_a^\kappa$ associated with FBM B^κ on an interval $[0, a]$ is defined as

$$\mathcal{H}_a^\kappa = \overline{\text{span}}\{B^\kappa(t), t \in [0, a]\}, \quad (6.3.3)$$

where the linear span

$$\text{span}\{B^\kappa(t), t \in [0, a]\} = \left\{ \sum_{j=1}^J c_j B^\kappa(t_j) : c_j \in \mathbb{R}, t_j \in [0, a], j = 1, \dots, J, J \geq 1 \right\} \quad (6.3.4)$$

consists of all possible linear combination of FBM on the interval $[0, a]$, and $\overline{\text{span}}$ indicates its closure in $L^2(\Omega)$.

Any linear combination of FBM on $[0, a]$ belongs to the Gaussian space $\mathcal{H}_a^\kappa$. Any finite collection $X_1, \dots, X_n$ of elements of $\mathcal{H}_a^\kappa$ forms a zero mean Gaussian vector $(X_1, \dots, X_n)$. This follows from the fact that Gaussianity is preserved under the $L^2(\Omega)$ -limits. The Gaussian space $\mathcal{H}_a^\kappa$ is a subset of $L^2(\Omega)$, and is naturally equipped with the $L^2(\Omega)$ inner product

$$(X, Y)_{\mathcal{H}_a^\kappa} = \mathbb{E}XY.$$

Since $\mathcal{H}_a^\kappa$ is defined as the closure in $L^2(\Omega)$, it is a *complete* inner product space; that is, a Hilbert space.⁴ Recall that an inner product space is complete if every Cauchy sequence of elements in the space has a limit that is also in the space. If the space is not complete, then the limit may either not exist or be outside the space (for example, it could be a so-called “generalized process”).

In view of (6.3.2), it is natural to try to associate the elements X of the Gaussian space $\mathcal{H}_a^\kappa$ with fractional Wiener integrals $\mathcal{I}_a^\kappa(f)$ and hence, more importantly, with *deterministic* functions or integrands f (and vice-versa). One natural association involves the set $\mathcal{E}$ of *elementary (step) functions* defined as

$$f(u) = \sum_{k=1}^n f_k 1_{(u_k, u_{k+1}]}(u), \quad (6.3.5)$$

where $f_k \in \mathbb{R}$ and $0 \leq u_k \leq u_{k+1} \leq a$, $k = 1, \dots, n$. For the elementary function f in (6.3.5), one naturally associates the random variable

$$\mathcal{I}_a^\kappa(f) = \sum_{k=1}^n f_k (B^\kappa(u_{k+1}) - B^\kappa(u_k)), \quad (6.3.6)$$

which is a finite combination of FBM increments. Conversely, any finite combination of FBM on the interval $[0, a]$ can be written as a fractional integral of an elementary function. Indeed, applying the summation by parts formula (A.2.5) in Appendix A.2 with $n = 1$, $a_k = 0$ for $k \geq m + 1$, yields

$$\sum_{k=1}^m c_k B^\kappa(u_k) = \widehat{C}_1 B^\kappa(u_1) + \sum_{k=2}^m \widehat{C}_k (B^\kappa(u_k) - B^\kappa(u_{k-1})) = \sum_{k=1}^m \widehat{C}_k (B^\kappa(u_k) - B^\kappa(u_{k-1})),$$

after we set $u_0 = 0$ and hence $B^\kappa(u_0) = 0$, where $\widehat{C}_k = \sum_{j=k}^m c_j$. For example, $3B^\kappa(1) + 2B^\kappa(2) = 5(B^\kappa(1) - B^\kappa(0)) + 2(B^\kappa(2) - B^\kappa(1)) = \mathcal{I}_a^\kappa(f)$ with $f(u) = 5(1_{(0,1]}(u)) + 2(1_{(1,2]}(u))$. In view of (6.3.4), one then has

$$\text{span}\{B^\kappa(t), t \in [0, a]\} = \{\mathcal{I}_a^\kappa(f), f \in \mathcal{E}\}$$

and by (6.3.3), we can take the closure in $L^2(\Omega)$ of both sides and get

$$\mathcal{H}_a^\kappa = \overline{\{\mathcal{I}_a^\kappa(f), f \in \mathcal{E}\}}.$$

This association between elementary functions (or their fractional Wiener integrals) and the elements of $\mathcal{H}_a^\kappa$ can naturally be extended to larger classes of functions $\mathcal{C}$ by using the following basic result. Note that a special attention is paid to whether $\mathcal{C}$ is complete. Some classes for FBM considered below will not be complete.

Proposition 6.3.2 *Let $\mathcal{E}$ be the set of elementary functions (6.3.5), $\mathcal{I}_a^\kappa$ be the map defined on elementary functions by (6.3.6) and let $\kappa \in (-1/2, 1/2)$. Suppose that $\mathcal{C}$ is a set of deterministic functions on $[0, a]$ such that*

⁴ The notation $\mathcal{H}$ in $\mathcal{H}_a^\kappa$ is chosen so as to suggest that the space is Hilbert. An inner product space is a linear (vector) space with an inner product.

- (1) $\mathcal{C}$ is an inner product space with an inner product $(f, g)_{\mathcal{C}}$ for $f, g \in \mathcal{C}$,
- (2) $\mathcal{E} \subset \mathcal{C}$ and $(f, g)_{\mathcal{C}} = \mathbb{E} \mathcal{I}_a^{\kappa}(f) \mathcal{I}_a^{\kappa}(g)$, for $f, g \in \mathcal{E}$,
- (3) the set $\mathcal{E}$ is dense in $\mathcal{C}$.

Then there is an isometry, denoted $\mathcal{I}_a^{\kappa}$, between the space $\mathcal{C}$ and a linear subspace of $\mathcal{H}_a^{\kappa}$ which is an extension of the map $f \rightarrow \mathcal{I}_a^{\kappa}(f)$, for $f \in \mathcal{E}$. Moreover, $\mathcal{C}$ is isometric to $\mathcal{H}_a^{\kappa}$ itself if and only if $\mathcal{C}$ is complete.

Proof To define the isometry $\mathcal{I}_a^{\kappa}$, let $f \in \mathcal{C}$. By the assumption (3), there is a sequence $\{f_n\} \subset \mathcal{E}$ such that $f_n \rightarrow f$ in $\mathcal{C}$. In particular, $\{f_n\}$ is Cauchy in $\mathcal{C}$ and hence, by the property (2), $\{\mathcal{I}_a^{\kappa}(f_n)\}$ is Cauchy in $L^2(\Omega)$. Since the space $L^2(\Omega)$ is complete, there is $X \in L^2(\Omega)$ such that $X = \lim \mathcal{I}_a^{\kappa}(f_n)$. Now, set $\mathcal{I}_a^{\kappa}(f) = X$. Since $\{\mathcal{I}_a^{\kappa}(f_n)\} \subset \mathcal{H}_a^{\kappa}$ and $\mathcal{H}_a^{\kappa}$ is a closed subset in $L^2(\Omega)$, we obtain that $\mathcal{I}_a^{\kappa}(f) \in \mathcal{H}_a^{\kappa}$. We can thus define the map $\mathcal{I}_a^{\kappa}$ from the space $\mathcal{C}$ into the space $\mathcal{H}_a^{\kappa}$. One can verify that this definition does not depend on the approximating sequence $\{f_n\}$ (Exercise 6.4). The construction of $\mathcal{I}_a^{\kappa}$ and the property (2) imply that, for $f, g \in \mathcal{C}$,

$$(f, g)_{\mathcal{C}} = \mathbb{E} \mathcal{I}_a^{\kappa}(f) \mathcal{I}_a^{\kappa}(g).$$

Moreover, since the map $\mathcal{I}_a^{\kappa}$ is linear, we conclude that $\mathcal{I}_a^{\kappa}$ is indeed an isometry between the space $\mathcal{C}$ and a linear subspace of $\mathcal{H}_a^{\kappa}$.

For the last statement of the proposition, if $\mathcal{C}$ is isometric to the full space $\mathcal{H}_a^{\kappa}$, then $\mathcal{C}$ is complete because the space $\mathcal{H}_a^{\kappa}$ is complete. Conversely, if $\mathcal{C}$ is complete, then the map is onto because $\mathcal{E}$ is dense in $\mathcal{C}$ and hence $\mathcal{C}$ is isometric to $\mathcal{H}_a^{\kappa}$ itself. $\square$

Definition 6.3.3 The isometry map $\mathcal{I}_a^{\kappa} : \mathcal{C} \mapsto \mathcal{H}_a^{\kappa}$ in Proposition 6.3.2 is denoted $\mathcal{I}_a^{\kappa}(f) = \int_0^a f(u) dB^{\kappa}(u)$, $f \in \mathcal{C}$. The latter integral is called a *fractional Wiener integral* of $f \in \mathcal{C}$, and $\mathcal{C}$ is called a *class of integrands*.

When $\kappa = 0$, FBM B^{κ} is the usual (standard) Brownian motion B^0 . For $f, g \in \mathcal{E}$, one can show (Exercise 6.5) that

$$\mathbb{E} \mathcal{I}_a^0(f) \mathcal{I}_a^0(g) = \int_0^a f(u) g(u) du. \quad (6.3.7)$$

A natural space of integrands for Brownian motion is then $\mathcal{C} = L^2[0, a]$, equipped with the usual inner product $(f, g)_{\mathcal{C}} = \int_0^a f(u) g(u) du$. The conditions (1)–(3) of Proposition 6.3.2 are satisfied. Since $L^2[0, a]$ is complete, it is isometric to $\mathcal{H}_a^0$. The resulting integral $\mathcal{I}_a^0(f) = \int_0^a f(u) dB^0(u)$, $f \in L^2[0, a]$, is called the *Wiener integral*. It can also be viewed as $\int_0^a f(u) B(du)$ where $B(du)$ is the Gaussian random measure with the control measure $\mathbb{E}|B(du)|^2 = du$.

Several classes $\mathcal{C}$ for FBM B^{κ} , $\kappa \neq 0$, are constructed in Section 6.3.2 below. The Gaussian space $\mathcal{H}_a^{\kappa}$ and fractional Wiener integrals are natural objects to use in a number of applications. For example, since B^{κ} is a Gaussian process, its predicted value at time $t \geq a$ given its past $B^{\kappa}(s)$, $s \in [0, a]$,

$$\mathbb{E} \left(B^{\kappa}(t) | B^{\kappa}(s), s \in [0, a] \right), \quad t \geq a, \quad (6.3.8)$$

is an element of $\mathcal{H}_a^\kappa$ and hence is expected to be written as a fractional Wiener integral. The corresponding prediction formula for (6.3.8) is derived in Section 6.4, which also includes other applications.

6.3.2 Classes of Integrands on an Interval

In this section, we construct classes of integrands for FBM on an interval $[0, a]$ with $a > 0$. Suppose that $\{B^\kappa(u)\}_{u \in [0, a]}$ is a standard FBM on an interval $[0, a]$. Then, by the representation (6.2.19), we obtain that, for every elementary function $f \in \mathcal{E}$ and $\kappa \in (-1/2, 1/2)$,

$$\mathcal{I}_a^\kappa(f) \stackrel{d}{=} \sigma_\kappa \int_0^a s^{-\kappa} \left(I_{a-}^\kappa u^\kappa f(u) \right)(s) dB^0(s), \quad (6.3.9)$$

where σ_κ is given in (6.2.20). It follows that, for all $f, g \in \mathcal{E}$,

$$\mathbb{E}(\mathcal{I}_a^\kappa(f) \mathcal{I}_a^\kappa(g)) = \sigma_\kappa^2 \int_0^a s^{-2\kappa} \left(I_{a-}^\kappa u^\kappa f(u) \right)(s) \left(I_{a-}^\kappa u^\kappa g(u) \right)(s) ds. \quad (6.3.10)$$

• **The case $\kappa \in (0, 1/2)$**

Consider first the case $\kappa \in (0, 1/2)$. Relations (6.3.9) and (6.3.10) lead us to introduce the following class of functions on an interval $[0, a]$

Definition 6.3.4 For $\kappa \in (0, 1/2)$, let

$$\Lambda_a^\kappa = \left\{ f : u^\kappa f(u) \in L^1[0, a] \text{ and } \int_0^a s^{-2\kappa} \left((I_{a-}^\kappa u^\kappa f(u))(s) \right)^2 ds < \infty \right\}. \quad (6.3.11)$$

We assume in (6.3.11) that the function $u^\kappa f(u)$ is in $L^1[0, a]$ so that the fractional integral $(I_{a-}^\kappa u^\kappa f(u))(s)$ is well-defined a.e. for $s \in [0, a]$ (see the discussion following Definition 6.1.1). The following result describes the space Λ_a^κ , and allows one to use Proposition 6.3.2 in constructing fractional Wiener integrals.

Theorem 6.3.5 For $\kappa \in (0, 1/2)$, the class of functions Λ_a^κ , defined by (6.3.11), is a linear space with the inner product

$$(f, g)_{\Lambda_a^\kappa} = \sigma_\kappa^2 \int_0^a s^{-2\kappa} \left(I_{a-}^\kappa u^\kappa f(u) \right)(s) \left(I_{a-}^\kappa u^\kappa g(u) \right)(s) ds. \quad (6.3.12)$$

The set of elementary functions $\mathcal{E}$ is dense in the space Λ_a^κ . The space Λ_a^κ is not complete.

Proof Note that $(f, f)_{\Lambda_a^\kappa} = 0$ implies $f = 0$ a.e. by Corollary 6.1.11. The other properties of an inner product, that is, linearity and symmetry, are also satisfied. Since Λ_a^κ is a linear space, it is then an inner product space. The fact that $\mathcal{E}$ is dense, is proved in Theorem 4.1 of Pipiras and Taqqu [815]. Since the latter fact is expected and the proof is technical, we will not provide it here. We will instead outline the steps to prove the last statement of the theorem concerning incompleteness.

First, let $0 \leq b \leq a$. As expected, the function

$$g_b(u) = u^{-\kappa} (I_{a-}^{-\kappa} s^\kappa 1_{[0, b)}(s))(u) = u^{-\kappa} (\mathcal{D}_{a-}^\kappa s^\kappa 1_{[0, b)}(s))(u) \quad (6.3.13)$$

satisfies the inverse relation

$$s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} g_b(u))(s) = 1_{[0,b)}(s) \quad (6.3.14)$$

(Exercise 6.6, (i)). Then, for $0 \leq c < b \leq a$, the function $g_{c,b}(u) = g_b(u) - g_c(u)$ satisfies

$$s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} g_{c,b}(u))(s) = 1_{[c,b)}(s). \quad (6.3.15)$$

Second, by using (6.3.15) and since linear combinations of indicator functions are dense in $L^2[0, a]$, one can show that Λ_a^{κ} is complete if and only if, for every $\phi \in L^2[0, a]$, there is a function $f_{\phi} \in \Lambda_a^{\kappa}$ such that

$$s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} f_{\phi}(u))(s) = \phi(s) \quad \text{a.e. } ds \quad (6.3.16)$$

(Exercise 6.6, (ii)). The basic idea now is that one cannot expect (6.3.16) to be true. Heuristically, since I_{a-}^{κ} is an operator which involves an integral, the left-hand side of (6.3.16) must satisfy some smoothness condition (for example, one can take its fractional derivative $(\mathcal{D}_{a-}^{\kappa} s^{\kappa})$ to obtain, by (6.1.13), $(\mathcal{D}_{a-}^{\kappa} s^{\kappa} s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} f_{\phi}(u))(s))(z) = (\mathcal{D}_{a-}^{\kappa} (I_{a-}^{\kappa} u^{\kappa} f_{\phi}(u)))(z) = z^{\kappa} f_{\phi}(z)$), whereas such condition need not to hold for a general $\phi \in L^2[0, a]$.

Third, one can actually provide an interesting example of a function $\phi \in L^2[0, a]$ for which (6.3.16) is not satisfied. In fact, we will argue first that there are continuous functions ψ on $[0, a]$ such that the equation

$$(I_{a-}^{\kappa} g(u))(s) = \psi(s) \quad \text{a.e. } ds \quad (6.3.17)$$

has no solution in $g \in L^1[0, a]$. Proceeding by contradiction, if (6.3.17) had a solution $g_{\psi} \in L^1[0, a]$, then by Proposition 6.1.8, (i), $g_{\psi}(u) = (\mathcal{D}_{a-}^{\kappa} \psi)(u)$. In particular, by the definition (6.1.12),

$$g_{\psi}(u) = (\mathcal{D}_{a-}^{\kappa} \psi)(u) = -\frac{1}{\Gamma(1-\kappa)} \frac{d}{du} \int_u^a \psi(s)(s-u)^{-\kappa} ds,$$

that is, the function

$$U_{\psi}(u) = \int_0^a \psi(s)(s-u)_+^{-\kappa} ds = \int_{\mathbb{R}} \psi(s+u) 1_{\{0 < s+u < a\}} s_+^{-\kappa} ds \quad (6.3.18)$$

would be differentiable a.e. du on $[0, a]$. We will show, for example, that when $a = 1$ (otherwise, use appropriate scaling), this is not the case for $U_{\psi^*}(u)$ where $\psi^*(s)$ is the Weierstrass function (see Section 3.13),

$$\psi^*(s) = \sum_{n=0}^{\infty} b^{-pn} e^{ib^n s}, \quad b > 1, \quad 0 < p < \kappa < 1/2. \quad (6.3.19)$$

The function ψ^* is complex-valued but one can focus on its real or imaginary part. As indicated in Section 3.13, these are continuous.

We want to show that $U_{\psi^*}(u)$ is not differentiable on $(0, 1)$. For $u \in (0, 1)$, we have by (6.3.18) with $a = 1$,

$$\begin{aligned}
U_{\psi^*}(u) &= \int_0^{1-u} s^{-\kappa} \left(\sum_{n=0}^{\infty} b^{-pn} e^{ib^n s} e^{ib^n u} \right) ds = \sum_{n=0}^{\infty} b^{-pn} \left(\int_0^{1-u} s^{-\kappa} e^{ib^n s} ds \right) e^{ib^n u} \\
&= \sum_{n=0}^{\infty} b^{-pn} \left(\int_0^{\infty} s^{-\kappa} e^{ib^n s} ds \right) e^{ib^n u} - \sum_{n=0}^{\infty} b^{-pn} \left(\int_{1-u}^{\infty} s^{-\kappa} e^{ib^n s} ds \right) e^{ib^n u} =: y_1(u) - y_2(u).
\end{aligned}$$

One can show that the function y_2 is differentiable on $[0, 1/2]$ (Pipiras and Taqqu [815], proof of Lemma 5.7). The function y_1 , on the other hand, is

$$y_1(u) = \left(\int_0^{\infty} z^{-\kappa} e^{iz} dz \right) \sum_{n=0}^{\infty} b^{-(p-\kappa+1)n} e^{ib^n u},$$

after the change of variables $b^n s = z$. Up to a multiplicative constant, since $b^{-(p-\kappa+1)}b = b^{\kappa-p} > 1$, y_1 is the Weierstrass function (3.13.1). It is nowhere differentiable on $[0, 1]$. We have thus constructed a continuous function ψ for which $U_{\psi}(u)$ is not differentiable, and hence (6.3.17) has no solution g .

Finally, and as a fourth point, if (6.3.17) does not have a solution for a function ψ , then after writing (6.3.17) as $s^{-\kappa}(I_{a-}^{\kappa}g(u))(s) = s^{-\kappa}\psi(s)$, (6.3.16) does not have a solution for $\phi(s) = s^{-\kappa}\psi(s)$. Moreover, since we can take a continuous function ψ as in (6.3.19), the function $\phi(s) = s^{-\kappa}\psi(s)$ is in $L^2[0, a]$ since $\kappa < 1/2$ and hence (6.3.16) does not have a solution for ϕ in $L^2[0, a]$. $\square$

Theorem 6.3.5 and Proposition 6.3.2 imply that the space Λ_a^{κ} of functions f is isometric to a *strict* subspace of $\mathcal{H}_a^{\kappa}$. Let $\mathcal{I}_a^{\kappa}$ denote the isometry map. We can then define the integral with respect to FBM as follows.

Definition 6.3.6 For $\kappa \in (0, 1/2)$ and $f \in \Lambda_a^{\kappa}$,

$$\mathcal{I}_a^{\kappa}(f) \equiv \int_0^a f(u) dB^{\kappa}(u) \stackrel{d}{=} \sigma_{\kappa} \int_0^a s^{-\kappa} \left(I_{a-}^{\kappa} u^{\kappa} f(u) \right)(s) dB^0(s), \quad (6.3.20)$$

where σ_{κ} is given in (6.2.20).

The space Λ_a^{κ} can be viewed as a class of deterministic integrands for FBM on an interval $[0, a]$. Note also that because of (6.3.11), we have:

Proposition 6.3.7 The covariance relation (6.3.10) is valid for $f, g \in \Lambda_a^{\kappa}$.

• **The case $\kappa \in (-1/2, 0)$**

Consider now the case $\kappa \in (-1/2, 0)$ and set $f(u) = u^{-\kappa}(I_{a-}^{\kappa} s^{\kappa} \phi(s))(u)$ for some function $\phi \in L^2[0, a]$. The function f is well-defined for $\phi \in L^2[0, a]$ because the function $s^{\kappa} \phi(s)$ is in $L^1[0, a]$ for $\kappa > -1/2$. Indeed, $\int_0^a |s^{\kappa} \phi(s)| ds \leq (\int_0^a s^{2\kappa} ds)^{1/2} (\int_0^a |\phi(s)|^2 ds)^{1/2} < \infty$ for $\kappa > -1/2$ by the Cauchy-Schwarz inequality. Property (i) of Proposition 6.1.8 then implies

$$s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} f(u))(s) = s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} u^{-\kappa} (I_{a-}^{\kappa} z^{\kappa} \phi(z))(u))(s) = \phi(s) \quad (6.3.21)$$

and hence that

$$\int_0^a s^{-2\kappa} \left((I_{a-}^{\kappa} u^{\kappa} f(u))(s) \right)^2 ds = \int_0^a \phi(s)^2 ds < \infty.$$

Based on this observation we introduce the class of functions:

Definition 6.3.8 For $\kappa \in (-1/2, 0)$, let

$$\Lambda_a^{\kappa} = \left\{ f : f(u) = u^{-\kappa} (I_{a-}^{-\kappa} s^{\kappa} \phi_f(s))(u) \text{ for } \phi_f \in L^2[0, a] \right\}. \quad (6.3.22)$$

Let us compare Definition 6.3.4 for $\kappa > 0$ to Definition 6.3.8 for $\kappa < 0$. In Definition 6.3.4, we require $u^{\kappa} f(u) \in L^1[0, a]$, but here we require that f results from a fractional integration involving a function $\phi_f \in L^2[0, a]$.

The next result is the analogue of Theorem 6.3.5 in the case $\kappa \in (-1/2, 0)$. We will not provide a proof. The fact that $\mathcal{E}$ is dense, is proved in Pipiras and Taqqu [817], Theorem 4.2. The completeness follows easily from the fact that the space $L^2[0, a]$ is complete (Pipiras and Taqqu [817], the discussion following Theorem 4.2).

Theorem 6.3.9 For $\kappa \in (-1/2, 0)$, the class of functions Λ_a^{κ} , defined by (6.3.22), is a linear space with the inner product

$$(f, g)_{\Lambda_a^{\kappa}} = \sigma_{\kappa}^2 \int_0^a s^{-2\kappa} (I_{a-}^{\kappa} u^{\kappa} f(u))(s) (I_{a-}^{\kappa} u^{\kappa} g(u))(s) ds \quad (6.3.23)$$

$$= \sigma_{\kappa}^2 \int_0^a \phi_f(s) \phi_g(s) ds, \quad (6.3.24)$$

where $\phi_f, \phi_g \in L^2[0, a]$ are associated with the functions f and g , respectively, by Definition 6.3.8. The set of elementary functions $\mathcal{E}$ is dense in the space Λ_a^{κ} . The space Λ_a^{κ} is complete.

Theorem 6.3.9 and Proposition 6.3.2 imply that the space Λ_a^{κ} is isometric to the space $\mathcal{H}_a^{\kappa}$ itself. If $\mathcal{I}_a^{\kappa}$ is the isometry map, we have (6.3.20) for $\kappa \in (-1/2, 0)$ as well. The space Λ_a^{κ} can be viewed as the class of deterministic integrands for FBM on an interval $[0, a]$. Definition 6.3.6 can thus be extended to $\kappa \in (-1/2, 0)$.

Definition 6.3.10 For $\kappa \in (-1/2, 0)$ and $f \in \Lambda_a^{\kappa}$,

$$\mathcal{I}_a^{\kappa}(f) \equiv \int_0^a f(u) dB^{\kappa}(u) \stackrel{d}{=} \sigma_{\kappa} \int_0^a s^{-\kappa} \left(I_{a-}^{\kappa} u^{\kappa} f(u) \right)(s) dB^0(s) = \sigma_{\kappa} \int_0^a \phi_f(s) dB^0(s). \quad (6.3.25)$$

Instead of first finding a function $\phi_f \in L^2[0, a]$ so that $f(u) = u^{-\kappa} (I_{a-}^{-\kappa} s^{\kappa} \phi_f(s))(u)$ and hence $s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} f(u))(s) = \phi_f(s)$, the next auxiliary result allows working with the (weighted) fractional derivative $s^{-\kappa} (I_{a-}^{\kappa} u^{\kappa} f(u))(s)$ directly. The result will sometimes be used to check that a function f belongs to the space Λ_a^{κ} and hence can be integrated with respect to FBM.

Lemma 6.3.11 Let $\kappa \in (-1/2, 0)$ and f be a function on $[0, a]$. Suppose that the (weighted) fractional derivative $s^{-\kappa}(I_{a-}^{\kappa} u^{\kappa} f(u))(s) =: \phi(s)$ can be evaluated directly. Then, if $\phi \in L^2[0, a]$, we have

$$u^{-\kappa}(I_{a-}^{-\kappa} s^{\kappa} \phi(s))(u) = f(u) \quad (6.3.26)$$

and hence $f \in \Lambda_a^{\kappa}$ with $\phi_f = \phi$.

Proof If $\phi \in L^2[0, a]$, we have $u^{-\kappa}(I_{a-}^{-\kappa} s^{\kappa} \phi(s))(u) = u^{-\kappa}(I_{a-}^{-\kappa} I_{a-}^{\kappa} v^{\kappa} f(v))(u) = f(u)$ by Proposition 6.1.8, (iii), as long as $(I_{a-}^{1+\kappa} v^{\kappa} f(v))$ is absolutely continuous. But since $(I_{a-}^{\kappa} u^{\kappa} f(u))(s)$ can be evaluated by definition, this means that the derivative of $\int_s^a u^{\kappa} f(u)(u-s)^{\kappa} du$ exists; that is, $(I_{a-}^{1+\kappa} v^{\kappa} f(v))(s)$ is differentiable (with the derivative $(I_{a-}^{\kappa} u^{\kappa} f(u))(s) = s^{\kappa} \phi(s)$, which is in $L^1[0, a]$) and hence is absolutely continuous. Thus, in this case, $f \in \Lambda_a^{\kappa}$ since ϕ_f can be taken as $\phi_f = \phi$. $\square$

The following result provides a useful and perhaps unexpected characterization of the space Λ_a^{κ} when $\kappa \in (-1/2, 0)$.

Proposition 6.3.12 For $\kappa \in (-1/2, 0)$,

$$\Lambda_a^{\kappa} = \left\{ f : f(u) = (I_{a-}^{-\kappa} \psi_f)(u) \text{ for } \psi_f \in L^2[0, a] \right\}. \quad (6.3.27)$$

Proof Consider the relation

$$u^{-\kappa}(I_{a-}^{-\kappa} s^{\kappa} \phi(s))(u) = (I_{a-}^{-\kappa} \psi)(u). \quad (6.3.28)$$

We need to show that, for $\phi \in L^2[0, a]$, there is $\psi \in L^2[0, a]$ satisfying (6.3.28), and vice versa. Thus, suppose $\phi \in L^2[0, a]$. Write the left-hand side of (6.3.28) as

$$\begin{aligned} f(u) &:= \Gamma(-\kappa) u^{-\kappa} (I_{a-}^{-\kappa} s^{\kappa} \phi(s))(u) = u^{-\kappa} \int_0^a s^{\kappa} \phi(s) (s-u)_{+}^{-\kappa-1} ds \\ &= u^{-\kappa} \int_{a^{-1}}^{\infty} (z^{-1})^{\kappa} \phi(z^{-1}) (z^{-1} - u)_{+}^{-\kappa-1} z^{-2} dz = u^{-2\kappa-1} \int_{a^{-1}}^{\infty} z^{-1} \phi(z^{-1}) (u^{-1} - z)_{+}^{-\kappa-1} dz \\ &= u^{-2\kappa-1} \int_{a^{-1}}^{\infty} z^{-1} \phi(z^{-1}) 1_{[a^{-1}, \infty)}(z) (u^{-1} - z)_{+}^{-\kappa-1} dz \\ &= \Gamma(-\kappa) u^{-2\kappa-1} (I_{0+}^{-\kappa} z^{-1} \phi(z^{-1}) 1_{[a^{-1}, \infty)}(z))(u^{-1}). \end{aligned} \quad (6.3.29)$$

Setting $f(u) = 0$ for $u > a$, the relation (6.3.29) holds for all $u > 0$ (not just $u < a$). Note that, since $\phi \in L^2[0, a]$, we have $z^{-1} \phi(z^{-1}) \in L^2[a^{-1}, \infty)$ and hence $z^{-1} \phi(z^{-1}) 1_{[a^{-1}, \infty)}(z) \in L^2(0, \infty)$. By Lemma 3.2 in Samko et al. [878], p. 70, since $\kappa > -1 + 1/2$, there is $g \in L^2(0, \infty)$ such that

$$(u^{-1})^{\kappa} (I_{0+}^{-\kappa} z^{-1} \phi(z^{-1}) 1_{[a^{-1}, \infty)}(z))(u^{-1}) = (I_{0+}^{-\kappa} z^{\kappa} g(z))(u^{-1}). \quad (6.3.30)$$

Since the left-hand side of (6.3.30) equals 0 for $u > a$ (or $u^{-1} < a^{-1}$), the right-hand side equals 0 for $u > a$ (or $u^{-1} < a^{-1}$) as well, and we can write (6.3.30) as

$$(u^{-1})^{\kappa} (I_{0+}^{-\kappa} z^{-1} \phi(z^{-1}) 1_{[a^{-1}, \infty)}(z))(u^{-1}) = (I_{0+}^{-\kappa} z^{\kappa} g(z) 1_{[a^{-1}, \infty)}(z))(u^{-1}). \quad (6.3.31)$$

Substituting (6.3.31) into (6.3.29) and working backwards in (6.3.29), we get that

$$\begin{aligned} f(u) &= \Gamma(-\kappa) u^{\kappa-1} (I_{0+}^{-\kappa} z^{\kappa} g(z) 1_{[a^{-1}, \infty)}(z))(u^{-1}) \\ &= u^{-\kappa-1} \int_{a^{-1}}^{\infty} z^{\kappa} g(z) (u^{-1} - z)_{+}^{-\kappa-1} dz = u^{-\kappa-1} \int_0^a (s^{-1})^{\kappa} g(s^{-1}) (u^{-1} - s^{-1})_{+}^{-\kappa-1} s^{-2} ds \\ &= \int_0^a s^{-1} g(s^{-1}) (s - u)_{+}^{-\kappa-1} ds = \Gamma(-\kappa) (I_{a-}^{-\kappa} s^{-1} g(s^{-1}))(u) =: \Gamma(-\kappa) (I_{a-}^{-\kappa} \psi)(u), \end{aligned}$$

where $\psi(s) = s^{-1} g(s^{-1})$. Since $g \in L^2[a^{-1}, \infty)$, we deduce that $\psi \in L^2[0, a]$, which proves (6.3.28) in one direction. The other direction can be proved analogously by working backwards in the argument above. $\square$

Remark 6.3.13 In view of (6.3.27), the space Λ_a^{κ} is often written as $\Lambda_a^{\kappa} = I_{a-}^{-\kappa}(L^2[0, a])$. The elements of the latter space can also be characterized as follows (Samko et al. [878], Theorem 13.4, p. 232): $f \in I_{a-}^{-\kappa}(L^2[0, a])$ if and only if

$$f(u)(u-a)^{\kappa} \in L^2[0, a] \quad \text{and} \quad \sup_{\epsilon > 0} \int_{\epsilon}^a \left| \int_a^{u-\epsilon} \frac{f(u) - f(s)}{(u-s)^{1-\kappa}} ds \right|^2 du < \infty. \quad (6.3.32)$$

6.3.3 Subspaces of Classes of Integrands

• The case $\kappa \in (0, 1/2)$

The class of integrands Λ_a^{κ} , $\kappa \in (0, 1/2)$, has a useful subspace. Suppose that the function f is such that $|f| \in \Lambda_a^{\kappa}$ so that $f \in \Lambda_a^{\kappa}$ as well. By using the relation (6.2.23) for $|u-v|^{2\kappa-1}$ and Fubini's theorem, we get that

$$\frac{B(\kappa, 1-2\kappa)}{\Gamma(\kappa)^2} \int_0^a \int_0^a f(u) f(v) |u-v|^{2\kappa-1} dudv = \int_0^a s^{-2\kappa} \left((I_{a-}^{\kappa} u^{\kappa} f(u))(s) \right)^2 ds. \quad (6.3.33)$$

Now consider the following space of functions:

Definition 6.3.14 For $\kappa \in (0, 1/2)$, set

$$|\Lambda|_a^{\kappa} = \left\{ f : \int_0^a \int_0^a |f(u)| |f(v)| |u-v|^{2\kappa-1} dudv < \infty \right\}. \quad (6.3.34)$$

By the relation (6.3.33), $|\Lambda|_a^{\kappa}$ is a linear subspace of Λ_a^{κ} . It also becomes an inner product space with the inner product $(f, g)_{|\Lambda|_a^{\kappa}}$ defined by $(f, g)_{\Lambda_a^{\kappa}}$ for $f, g \in |\Lambda|_a^{\kappa}$ and hence can also be viewed as a class of integrands for FBM. By using the relation (6.2.23) for $|u-v|^{2\kappa-1}$ and Fubini's theorem, we have that

$$(f, g)_{|\Lambda|_a^{\kappa}} = \kappa(2\kappa + 1) \int_0^a \int_0^a f(u) g(v) |u-v|^{2\kappa-1} dudv. \quad (6.3.35)$$

Since for $\kappa \in (0, 1/2)$, Λ_a^{κ} is not complete, the space $|\Lambda|_a^{\kappa}$ is not complete either (Pipiras and Taqqu [817], Theorem 5.1). We restate (6.3.35) as the following useful result.

Proposition 6.3.15 If $\kappa \in (0, 1/2)$ and $f, g \in |\Lambda|_a^\kappa$, then the covariance structure of the integrals $\mathcal{I}_a^\kappa(f)$ and $\mathcal{I}_a^\kappa(g)$ can also be expressed as

$$\mathbb{E}\mathcal{I}_a^\kappa(f)\mathcal{I}_a^\kappa(g) = \kappa(2\kappa + 1) \int_0^a \int_0^a f(u)g(v)|u - v|^{2\kappa-1} du dv.$$

The function space $|\Lambda|_a^\kappa$ is sometimes more convenient to work with than the space Λ_a^κ . For example, one may define on $|\Lambda|_a^\kappa$ a norm $\|\cdot\|_a$ by

$$\|f\|_a^2 = \int_0^a \int_0^a |f(u)||f(v)||u - v|^{2\kappa-1} du dv, \quad (6.3.36)$$

which is easier to manipulate than the norm induced by the inner product on Λ_a^κ ; that is,

$$\|f\|_{|\Lambda|_a^\kappa}^2 = \int_0^a \int_0^a f(u)f(v)|u - v|^{2\kappa-1} du dv. \quad (6.3.37)$$

Moreover, as the next proposition states, the space $|\Lambda|_a^\kappa$ is complete under the new norm $\|\cdot\|_a$.

Proposition 6.3.16 The space $(|\Lambda|_a^\kappa, \|\cdot\|_a)$ when $\kappa \in (0, 1/2)$ is a complete normed space.

Proof The fact that $\|\cdot\|_a$ is a norm is elementary. To show completeness, let $\{f_n\}$ be a Cauchy sequence in $(|\Lambda|_a^\kappa, \|\cdot\|_a)$, that is, $\|f_n - f_m\|_a \rightarrow 0$, as $n, m \rightarrow \infty$. We have to find $f \in |\Lambda|_a^\kappa$ such that $\|f - f_n\|_a \rightarrow 0$, as $n \rightarrow \infty$. For this, observe that, since $|u - v|^{2\kappa-1} \geq a^{2\kappa-1}$ for $u, v \in (0, a)$, $\|g\|_a^2 \geq a^{2\kappa-1} \int_0^a \int_0^a |g(u)||g(v)| du dv = a^{2\kappa-1} (\int_0^a |g(u)| du)^2$. Thus, $\{f_n\}$ is also a Cauchy sequence in $L^1[0, a]$. Since the space $L^1[0, a]$ is complete, there is a function $f \in L^1[0, a]$ such that $f_n \rightarrow f$ in $L^1[0, a]$. There is also a subsequence n_l such that $f_{n_l} \rightarrow f$ a.e. To show that f is the desired limit in $\|\cdot\|_a$, note that, by Fatou's lemma, that $\|f\|_a = \|\liminf_l f_{n_l}\|_a \leq \liminf_l \|f_{n_l}\|_a < \infty$. Then, by Fatou's lemma again, $\|f - f_n\|_a = \|\liminf_l f_{n_l} - f_n\|_a \leq \liminf_l \|f_{n_l} - f_n\|_a \rightarrow 0$, as $n \rightarrow \infty$, since $\{f_n\}$ is Cauchy in the norm $\|\cdot\|_a$. $\square$

The following basic result provides other useful subsets of the spaces introduced.

Proposition 6.3.17 For $\kappa \in (0, 1/2)$,

$$L^2[0, a] \subset L^{2/(2\kappa+1)}[0, a] \subset |\Lambda|_a^\kappa \subset \Lambda_a^\kappa. \quad (6.3.38)$$

Proof The inclusion $L^2[0, a] \subset L^{2/(2\kappa+1)}[0, a]$ is obvious, since $2/(2\kappa + 1) < 2$ for $\kappa > 0$. To show the inclusion $L^{2/(2\kappa+1)}[0, a] \subset |\Lambda|_a^\kappa$, observe that, by Hölder's inequality,

$$\begin{aligned} \|f\|_a^2 &= \int_0^a \int_0^a |f(u)||f(v)||u - v|^{2\kappa-1} du dv = \int_0^a |f(u)| \left(\int_0^a |f(v)||u - v|^{2\kappa-1} dv \right) du \\ &\leq \left(\int_0^a |f(u)|^{\frac{2}{2\kappa+1}} du \right)^{\frac{2\kappa+1}{2}} \left(\int_0^a \left(\int_0^a |f(v)||u - v|^{2\kappa-1} dv \right)^{\frac{2}{1-2\kappa}} du \right)^{\frac{1-2\kappa}{2}} \\ &= \Gamma(2\kappa) \left(\int_0^a |f(u)|^{\frac{2}{2\kappa+1}} du \right)^{\frac{2\kappa+1}{2}} \left(\int_0^a \left((I_{0+}^{2\kappa}|f|)(u) + (I_{a-}^{2\kappa}|f|)(u) \right)^{\frac{2}{1-2\kappa}} du \right)^{\frac{1-2\kappa}{2}}. \end{aligned}$$

By using Theorem 3.5 in Samko et al. [878], p. 66, $I_{0+}^{2\kappa}$ and $I_{a-}^{2\kappa}$ are bounded operators from $L^{2/(2\kappa+1)}[0, a]$ to $L^{2/(1-2\kappa)}[0, a]$. This establishes the inclusion. $\square$

Remark 6.3.18 The end of the proof above also yields the following useful inequalities: for $\kappa \in (0, 1/2)$, there are constants C, C' such that

$$\|f\|_a \leq C\|f\|_{L^{2/(2\kappa+1)}[0,a]} \leq C'\|f\|_{L^2[0,a]}. \quad (6.3.39)$$

• **The case** $\kappa \in (-1/2, 0)$

When $\kappa \in (-1/2, 0)$, a useful subspace of Λ_a^κ is specified in the following result. Let

$$C^\gamma[0, a] = \{f : [0, a] \rightarrow \mathbb{R} : |f(u) - f(v)| \leq C|u - v|^\gamma\} \quad (6.3.40)$$

be the space of γ -Hölder continuous functions on $[0, a]$.

Proposition 6.3.19 Let $\kappa \in (-1/2, 0)$ and $0 < (-\kappa) < \gamma$. Then, for $f \in C^\gamma[0, a]$,

$$s^{-\kappa}(I_{a-}^\kappa u^\kappa f(u))(s) = \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))u^\kappa (u - s)_+^{\kappa-1} du + s^{-\kappa} f(s)(I_{a-}^\kappa u^\kappa)(s) \quad (6.3.41)$$

$$\begin{aligned} &= \frac{1}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))(u - s)_+^{\kappa-1} du + \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a f(u)(u^\kappa - s^\kappa)(u - s)_+^{\kappa-1} du \\ &\quad + \frac{f(s)(a - s)^\kappa}{\Gamma(\kappa)} \end{aligned} \quad (6.3.42)$$

and

$$C^\gamma[0, a] \subset \Lambda_a^\kappa. \quad (6.3.43)$$

Remark 6.3.20 Informally, the relation (6.3.41) follows from

$$\begin{aligned} -\Gamma(\kappa + 1)(I_{a-}^\kappa u^\kappa f(u))(s) &= \frac{d}{ds} \int_0^a f(u)u^\kappa (u - s)_+^\kappa du = -\kappa \int_0^a f(u)u^\kappa (u - s)_+^{\kappa-1} du \\ &= -\kappa \int_0^a (f(u) - f(s))u^\kappa (u - s)_+^{\kappa-1} du - \kappa f(s) \int_0^a u^\kappa (u - s)_+^{\kappa-1} du \\ &= -\kappa \int_0^a (f(u) - f(s))u^\kappa (u - s)_+^{\kappa-1} du + f(s) \frac{d}{ds} \int_0^a u^\kappa (u - s)_+^\kappa du. \end{aligned}$$

Proof We first show the equality of the right-hand sides of (6.3.41) and (6.3.42). Observe that the right-hand side of (6.3.41) is equal to

$$\begin{aligned} &\frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))s^\kappa (u - s)_+^{\kappa-1} du + \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))(u^\kappa - s^\kappa)(u - s)_+^{\kappa-1} du \\ &+ s^{-\kappa} f(s)(I_{a-}^\kappa u^\kappa)(s) = \frac{1}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))(u - s)_+^{\kappa-1} du \\ &+ \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a f(u)(u^\kappa - s^\kappa)(u - s)_+^{\kappa-1} du \\ &- \frac{f(s)s^{-\kappa}}{\Gamma(\kappa)} \int_0^a (u^\kappa - s^\kappa)(u - s)_+^{\kappa-1} du + s^{-\kappa} f(s)(I_{a-}^\kappa u^\kappa)(s) \end{aligned}$$

$$= \frac{1}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))(u-s)_+^{\kappa-1} du + \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a f(u)(u^\kappa - s^\kappa)(u-s)_+^{\kappa-1} du \\ + \frac{f(s)(a-s)^\kappa}{\Gamma(\kappa)}$$

by using (6.2.15) with $t = a$ in the last step, which leads to the right-hand side of (6.3.42).

We now prove (6.3.42). Let $g(s) = \int_0^a f(u)u^\kappa(u-s)_+^{\kappa-1} du$. We need to show that g is differentiable and that

$$\frac{d}{ds}g(s) = \frac{1}{\Gamma(\kappa)} \int_0^a (f(u) - f(s))(u-s)_+^{\kappa-1} du + \frac{s^{-\kappa}}{\Gamma(\kappa)} \int_0^a f(u)(u^\kappa - s^\kappa)(u-s)_+^{\kappa-1} du \\ + \frac{f(s)(a-s)^\kappa}{\Gamma(\kappa)} =: I_1 + I_2 + I_3. \quad (6.3.44)$$

For example, for $h > 0$, observe that

$$g(s+h) - g(s) = \int_{s+h}^a f(u)u^\kappa(u-s-h)^\kappa du - \int_s^a f(u)u^\kappa(u-s)^\kappa du \\ = \int_{s+h}^a f(u)u^\kappa \left((u-s-h)^\kappa - (u-s)^\kappa \right) du - \int_s^{s+h} f(u)u^\kappa(u-s)^\kappa du \\ = -\kappa \int_{s+h}^a f(u)u^\kappa \left(\int_s^{s+h} (u-z)^{\kappa-1} dz \right) du - \int_s^{s+h} f(u)u^\kappa(u-s)^\kappa du \\ = -\kappa \int_s^{s+h} \left(\int_{s+h}^a f(u)u^\kappa(u-z)^{\kappa-1} du \right) dz - \int_s^{s+h} f(u)u^\kappa(u-s)^\kappa du \\ = -\kappa \int_s^{s+h} \left(\int_{s+h}^a (f(u) - f(z))(u-z)^{\kappa-1} du \right) z^\kappa dz \\ - \kappa \int_s^{s+h} \left(\int_{s+h}^a f(u)(u^\kappa - z^\kappa)(u-z)^{\kappa-1} du \right) dz \\ - \kappa \int_s^{s+h} f(z)z^\kappa \left(\int_{s+h}^a (u-z)^{\kappa-1} du \right) dz - \int_s^{s+h} f(u)u^\kappa(u-s)^\kappa du \\ = -\kappa \int_s^{s+h} \left(\int_{s+h}^a (f(u) - f(z))(u-z)^{\kappa-1} du \right) z^\kappa dz \\ - \kappa \int_s^{s+h} \left(\int_{s+h}^a f(u)(u^\kappa - z^\kappa)(u-z)^{\kappa-1} du \right) dz \\ - \int_s^{s+h} f(z)z^\kappa(a-z)^\kappa dz + \int_s^{s+h} f(z)z^\kappa(s+h-z)^\kappa dz \\ - \int_s^{s+h} f(u)u^\kappa(u-s)^\kappa du =: I_{1,h} + I_{2,h} + I_{3,h} + I_{4,h} + I_{5,h}. \quad (6.3.45)$$

To prove (6.3.44), it is enough to show that $h^{-1}I_{k,h} \rightarrow I_k$ for $k = 1, 2, 3$ as $h \rightarrow 0$, and $h^{-1}(I_{4,h} + I_{5,h}) \rightarrow 0$ as $h \rightarrow 0$. This is left as Exercise 6.7.

Consider now (6.3.43) and take $f \in C^\gamma[0, a]$. We need to show that $f \in \Lambda_a^\kappa$, that is, $s^{-\kappa}(I_{a-}^\kappa u^\kappa f(u))(s)$ or the right-hand side of (6.3.41) belongs to $L^2[0, a]$ (see Lemma

6.3.11). By using the bounds $|f(u) - f(v)| \leq C|u - v|^\gamma$ in the first term of (6.3.41) and $|f(s)| \leq C$ in the second term of (6.3.41), it is enough to prove that $g_1(s), g_2(s) \in L^2[0, a]$, where

$$g_1(s) = s^{-\kappa} \int_0^a u^\kappa (u - s)_+^{\gamma+\kappa-1} du, \quad g_2(s) = s^{-\kappa} (I_{a-}^\kappa u^\kappa)(s).$$

The function $g_2(s)$ equals the kernel function $f_t(s)$ of FBM at $t = a$ (see (6.2.3)) and hence belongs to $L^2[0, a]$. To see that $g_1 \in L^2[0, a]$, write g_1 as, after the change of variables $u = sv$,

$$g_1(s) = s^{\kappa+\gamma} \int_1^{a/s} v^\kappa (v - 1)^{\gamma+\kappa-1} dv. \quad (6.3.46)$$

The function $g_1(s)$ is continuous for $s \in (0, a]$. Since a γ -Hölder function is also γ' -Hölder for $\gamma' < \gamma$, we may assume without loss of generality that $\gamma + 2\kappa < 0$ or $\gamma < (-2\kappa)$. In addition, by assumption $\gamma + \kappa > 0$. Then, the integral in (6.3.46) converges and therefore $g_1(s) \sim Cs^{\kappa+\gamma}$ as $s \rightarrow 0$. Hence g_1 belongs to $L^2[0, a]$. $\square$

Remark 6.3.21 Let $\kappa \in (-1/2, 0)$. By Theorem 3.5 in Samko et al. [878], p. 55, since $1 < 2 < 1/(-\kappa)$,

$$\Lambda_a^\kappa = I_{a-}^{-\kappa}(L^2[0, a]) \subset L^q[0, a] \subset L^2[0, a], \quad (6.3.47)$$

where $q = 2/(1 + 2\kappa) > 2$. By Theorem 14.2 in Samko et al. [878], p. 256, if $f \in \Lambda_a^\kappa = I_{a-}^{-\kappa}(L^2[0, a])$, then

$$w_2(f, \epsilon) = \sup_{|t| \leq \epsilon} \left(\int_0^a |f(u) - f(u - t)|^2 du \right)^{1/2} \leq C\epsilon^{-\kappa}, \quad (6.3.48)$$

where one sets $f(u - t) = 0$ when $u - t < 0$. This result is useful in showing that a function, which fails to satisfy (6.3.48) and hence is not regular enough, does not belong to the space of integrands Λ_a^κ .

In the following sections, we provide several examples involving fractional Wiener integrals.

6.3.4 The Fundamental Martingale

Let $\{B^\kappa(u)\}_{u \in [0, a]}$ be a standard FBM with parameter $\kappa \in (-1/2, 1/2)$. For $t \in [0, a]$, define

$$M^\kappa(t) = \frac{1}{\sigma_{2,\kappa}} \int_0^a u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0,t]})(u) dB^\kappa(u), \quad (6.3.49)$$

where $\sigma_{2,\kappa} = (2\kappa + 1)\Gamma(1 + \kappa)$ is a constant chosen for convenience (see Lemma 6.4.1 below). Observe that M^κ can be written as

$$M^\kappa(t) = \frac{1}{\sigma_{2,\kappa}\Gamma(1 - \kappa)} \int_0^t u^{-\kappa} (t - u)^{-\kappa} dB^\kappa(u), \quad (6.3.50)$$

since $\Gamma(1 - \kappa)(I_{a-}^{-\kappa} 1_{[0,t]})(u) = (t - u)_+^{-\kappa}$. When $(-\kappa) > 0$, the last identity follows from $\Gamma(1 - \kappa)(I_{a-}^{-\kappa} 1_{[0,t]})(u) = \Gamma(1 - \kappa)(I_{-}^{-\kappa} 1_{[0,t]})(u) = (t - u)_+^{-\kappa} - (-u)_+^{-\kappa} = (t - u)_+^{-\kappa}$ for

$u > 0$, where we used (6.1.24); when $(-\kappa) < 0$, this follows from $\Gamma(1-\kappa)(I_{a-}^{-\kappa} 1_{[0,t)})(u) = \Gamma(1-\kappa)(\mathcal{D}_{a-}^{\kappa} 1_{[0,t)})(u) = \Gamma(1-\kappa)(\mathcal{D}_{a-}^{-\kappa} 1_{[0,t)})(u) = (t-u)_{+}^{-\kappa} - (-u)_{+}^{-\kappa} = (t-u)_{+}^{-\kappa}$ for $u > 0$, where we used (6.1.25). Then, by the definition of the integral with respect to FBM B^{κ} (Definitions 6.3.6 and 6.3.10), we get that

$$M^{\kappa}(t) \stackrel{d}{=} \frac{\sigma_{\kappa}}{\sigma_{2,\kappa}} \int_0^t s^{-\kappa} \left(I_{a-}^{\kappa} u^{\kappa} u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0,t)})(u) \right)(s) dB^0(s) = \frac{\sigma_{\kappa}}{\sigma_{2,\kappa}} \int_0^t s^{-\kappa} dB^0(s), \quad (6.3.51)$$

so that the process M^{κ} is an integral with respect to Brownian motion and hence a Gaussian martingale with the quadratic variation⁵

$$[M^{\kappa}, M^{\kappa}]_t = \mathbb{E}(M^{\kappa}(t)^2) = \frac{\sigma_{\kappa}^2}{\sigma_{2,\kappa}^2} \int_0^t s^{-2\kappa} ds = \frac{\sigma_{\kappa}^2}{\sigma_{2,\kappa}^2(1-2\kappa)} t^{1-2\kappa}. \quad (6.3.52)$$

where the first equality follows from the facts that $\{M^{\kappa}(t)^2 - [M^{\kappa}, M^{\kappa}]_t\}_{t \in [0,a]}$ is a martingale and $[M^{\kappa}, M^{\kappa}]_0 = 0$ (see, e.g., Klebaner [562], Theorem 7.27, p. 199), and hence that $\mathbb{E}(M^{\kappa}(t)^2 - [M^{\kappa}, M^{\kappa}]_t) = \mathbb{E}(M^{\kappa}(0)^2 - [M^{\kappa}, M^{\kappa}]_0) = 0$. Similarly, if $t_1 < t_2$,

$$\mathbb{E}M^{\kappa}(t_1)M^{\kappa}(t_2) = \frac{\sigma_{\kappa}^2}{\sigma_{2,\kappa}^2(1-2\kappa)} t_1^{1-2\kappa}.$$

Hence $M^{\kappa}(t)$ is nothing else than

$$\{M^{\kappa}(t)\} \stackrel{d}{=} \left\{ \frac{\sigma_{\kappa}}{\sigma_{2,\kappa}(1-2\kappa)^{1/2}} B^0(t^{1-2\kappa}) \right\},$$

that is, Brownian motion with rescaled time $t \rightarrow t^{1-2\kappa}$. The process M^{κ} is called the *fundamental martingale* because it plays an important role, for example, in Girsanov's formula (Section 6.4.1).

6.3.5 The Deconvolution Formula

Set

$$f_{\kappa}(u, t) = u^{-\kappa} (I_{a-}^{-\kappa} s^{\kappa} 1_{[0,t)}(s))(u), \quad u, t \in [0, a],$$

for $\kappa \in (-1/2, 1/2)$. We will first show that $f_{\kappa}(\cdot, t) \in \Lambda_a^{\kappa}$, where Λ_a^{κ} is given in Definition 6.3.4 when $\kappa \in (0, 1/2)$ and in Definition 6.3.8 when $\kappa \in (-1/2, 0)$. We will then establish a “deconvolution formula” for FBM.

Arguing as for (6.2.8), one can verify that

$$-\Gamma(1-\kappa)f_{\kappa}(u, t) = \kappa u^{-\kappa} \int_u^t z^{\kappa-1} (z-u)_{+}^{-\kappa} dz - \left(\frac{t}{u}\right)^{\kappa} (t-u)_{+}^{-\kappa}, \quad (6.3.53)$$

⁵ Quadratic variation of a process $X(t)$ is defined as a limit in probability

$$[X, X]_t = \lim \sum_{i=1}^n (X(t_i^n) - X(t_{i-1}^n))^2,$$

where the limit is taken over partitions $0 = t_0^n < t_1^n < \dots < t_n^n = t$ with $\max_{1 \leq i \leq n} (t_i^n - t_{i-1}^n) \rightarrow 0$.

for $u, t \in [0, a]$. When $\kappa \in (-1/2, 0)$, since $1_{[0,t)} \in L^2[0, a]$, we obtain from (6.3.22) that $f_\kappa(\cdot, t) \in \Lambda_a^\kappa$ and from (6.3.21) that

$$s^{-\kappa} (I_{a-}^\kappa u^\kappa f_\kappa(u, t))(s) = 1_{[0,t)}(s), \quad s \in [0, a]. \quad (6.3.54)$$

When $\kappa \in (0, 1/2)$, the relation (6.3.54) holds as well. To see this, observe that, by using (6.1.3), (6.3.53), the relation $\int_c^d (u-c)^{\kappa-1} (d-u)^{-\kappa} du = B(1-\kappa, \kappa) = \Gamma(\kappa)\Gamma(1-\kappa)$ with $c < d$ (see (6.1.9)) and Fubini's theorem,

$$\begin{aligned} & -\Gamma(\kappa)\Gamma(1-\kappa)(I_{a-}^\kappa u^\kappa f_\kappa(u, t))(s) \\ &= \kappa \int_s^t (u-s)_+^{\kappa-1} \int_u^t z^{\kappa-1} (z-u)_+^{-\kappa} dz du - t^\kappa \int_s^t (u-s)_+^{\kappa-1} (t-u)_+^{-\kappa} du \\ &= \kappa \int_s^t dz z^{\kappa-1} \int_s^z du (u-s)_+^{\kappa-1} (z-u)_+^{-\kappa} - t^\kappa B(1-\kappa, \kappa) 1_{[0,t)}(s) \\ &= B(1-\kappa, \kappa) 1_{[0,t)}(s) \left(z^\kappa \Big|_{z=s}^t - t^\kappa \right) = -\Gamma(\kappa)\Gamma(1-\kappa) s^\kappa 1_{[0,t)}(s). \end{aligned}$$

Since $1_{[0,t)} \in L^2[0, a]$, we also obtain from (6.3.54) and (6.3.11) that $f_\kappa(\cdot, t) \in \Lambda_a^\kappa$ when $\kappa \in (0, 1/2)$.

It follows from (6.3.20), (6.3.25) and (6.3.54) that, when $\kappa \in (-1/2, 1/2)$,

$$\begin{aligned} \left\{ (\sigma_\kappa)^{-1} \int_0^a f_\kappa(u, t) dB^\kappa(u) \right\}_{t \in [0, a]} &\stackrel{d}{=} \left\{ \int_0^a s^{-\kappa} (I_{a-}^\kappa u^\kappa f_\kappa(u, t))(s) dB^0(s) \right\}_{t \in [0, a]} \\ &= \left\{ \int_0^a 1_{[0,t)}(s) dB^0(s) \right\}_{t \in [0, a]} = \{B^0(t)\}_{t \in [0, a]}. \end{aligned} \quad (6.3.55)$$

The expression involving f_κ in (6.3.55) is the *deconvolution formula* for FBM on an interval $[0, a]$. It compensates for the dependence of FBM and yields Brownian motion, a process with independent increments. It is used for example in deriving the prediction formula for FBM (see Section 6.4.2).

6.3.6 Classes of Integrands on the Real Line

Fractional Wiener integrals can similarly be defined on the real line. Denoted

$$\mathcal{I}^\kappa(f) = \int_{\mathbb{R}} f(u) dB^\kappa(u) \quad (6.3.56)$$

for deterministic functions f on the real line, these integrals are now elements of the Gaussian space $\overline{\text{span}}\{B^\kappa(t), t \in \mathbb{R}\}$ (cf. Definition 6.3.1). Several classes of integrands f can be defined, including

$$\Lambda^\kappa = \begin{cases} \{f : \int_{\mathbb{R}} ((I_-^\kappa f)(s))^2 ds < \infty\}, & \text{if } \kappa \in (0, 1/2), \\ \{f : f = I_-^\kappa \phi_f\} \text{ for } \phi_f \in L^2(\mathbb{R}), & \text{if } \kappa \in (-1/2, 0), \end{cases} \quad (6.3.57)$$

$$|\Lambda|^\kappa = \left\{ f : \int_{\mathbb{R}} \int_{\mathbb{R}} |f(u)| |f(v)| |u-v|^{2\kappa-1} dudv < \infty \right\}, \quad \text{if } \kappa \in (0, 1/2), \quad (6.3.58)$$

and also

$$\widehat{\Lambda}^\kappa = \left\{ f : f \in L^2(\mathbb{R}), \int_{\mathbb{R}} |\widehat{f}(x)| |x|^{-2\kappa} dx < \infty \right\}, \quad \text{if } \kappa \in (-1/2, 1/2). \quad (6.3.59)$$

The following inclusions hold for the classes of integrands (6.3.57)–(6.3.59):

$$\widehat{\Lambda}^\kappa \subset \Lambda^\kappa, \quad \kappa \in (-1/2, 1/2), \quad (6.3.60)$$

$$L^1(\mathbb{R}) \cap L^2(\mathbb{R}) \subset L^{2/(2\kappa+1)}(\mathbb{R}) \subset |\Lambda|^\kappa \subset \Lambda^\kappa, \quad \kappa \in (0, 1/2). \quad (6.3.61)$$

For example, the construction of the class Λ^κ is based on the representation (6.2.29) of a standard FBM, and the corresponding fractional Wiener integrals are defined as

$$\mathcal{I}^\kappa(f) = \int_{\mathbb{R}} f(u) dB^\kappa(u) \stackrel{d}{=} \frac{\Gamma(\kappa+1)}{c_{1,\kappa}} \int_{\mathbb{R}} (I_-^\kappa f)(s) dB^0(s), \quad (6.3.62)$$

for $f \in \Lambda^\kappa$, $\kappa \in (-1/2, 1/2)$. For $\kappa \in (-1/2, 0)$, the last integral is also $\int_{\mathbb{R}} \phi_f(s) dB^0(s)$. See Pipiras and Taqqu [815] for more information on fractional Wiener integrals on the real line.

Example 6.3.22 In analogy to the Ornstein–Uhlenbeck process in Examples 1.1.2 and 1.4.3, consider the process

$$Y^\kappa(t) = \int_{\mathbb{R}} e^{-\lambda(t-u)} 1_{\{t-u>0\}} dB^\kappa(u) = e^{-\lambda t} \int_{\mathbb{R}} e^{\lambda u} 1_{\{u<t\}} dB^\kappa(u), \quad (6.3.63)$$

where $\lambda > 0$ and $\kappa \in (-1/2, 1/2)$. The process is well-defined (for fixed t) since the integrand $f_t(u) = e^{\lambda u} 1_{\{u<t\}}$ belongs to $\widehat{\Lambda}^\kappa \subset \Lambda^\kappa$. Indeed, its Fourier transform is

$$\widehat{f}_t(x) = \int_{\mathbb{R}} e^{ixu} e^{\lambda u} 1_{\{u<t\}} du = \frac{e^{(\lambda+ix)t}}{\lambda+ix}$$

and hence $|\widehat{f}_t(x)|^2 = e^{\lambda t}/(\lambda^2 + x^2)$ satisfies $\int_{\mathbb{R}} |\widehat{f}_t(x)| |x|^{-2\kappa} dx < \infty$. The process (6.3.63) is called the fractional Ornstein–Uhlenbeck (OU) process. It is zero mean, Gaussian and stationary.

The fractional OU process satisfies the Langevin equation

$$Y^\kappa(t) = \xi - \lambda \int_0^t Y^\kappa(s) ds + B^\kappa(t), \quad t \geq 0, \quad (6.3.64)$$

with FBM B^κ as the driving process (cf. Example 1.1.2) and $\xi = Y^\kappa(0)$. The equation (6.3.64) and the fractional OU process can be interpreted in the pathwise sense (Cheridito, Kawaguchi, and Maejima [229]). In a weaker and simpler sense, one could verify that the fractional OU process satisfies (6.3.64) for fixed t . The latter and other basic properties of the fractional OU process can be found in Exercise 6.8.

6.3.7 Connection to the Reproducing Kernel Hilbert Space

The reproducing kernel Hilbert space (RKHS) of FBM B^κ provides an alternative characterization of the Gaussian space $\mathcal{H}_a^\kappa$ defined in (6.3.3). We first recall the definition of a RKHS (Grenander [429], p. 93, Weinert [1002]). Suppose that $X = \{X(t)\}_{t \in T}$ is a zero mean

stochastic process indexed by a set T with the covariance function $\Gamma(t_1, t_2)$, $t_1, t_2 \in T$, and let $\overline{\text{span}}(X)$ be the closure in $L^2(\Omega)$ of all linear combinations $\sum_{i=1}^k a_i X(t_i)$, $a_i \in \mathbb{R}$, $t_i \in T$. Then with the Hilbert space $\overline{\text{span}}(X)$ one can always associate an isometric Hilbert space $\mathbb{H}(\Gamma)$ of deterministic functions. The space $\mathbb{H}(\Gamma)$ is called the RKHS and is characterized by the following two properties:

- (1) $\Gamma(\cdot, t) \in \mathbb{H}(\Gamma)$, for all $t \in T$, and
- (2) $(g, \Gamma(\cdot, t))_{\mathbb{H}(\Gamma)} = g(t)$, for all $t \in T$ and $g \in \mathbb{H}(\Gamma)$.

It consists of all functions of the form $f(t) = \sum_{i=1}^k a_i \Gamma(t, t_i)$, $t \in T$, and their limits under the norm $\|f\|^2 = \sum_{i,j=1}^k a_i a_j \Gamma(t_i, t_j)$. The isometry map $\mathcal{J}$ between the Hilbert spaces $\overline{\text{span}}(X)$ and $\mathbb{H}(\Gamma)$ satisfies

$$\mathcal{J} : \sum_{i=1}^k a_i \Gamma(\cdot, t_i) \mapsto \sum_{i=1}^k a_i X(t_i) \quad (6.3.65)$$

and $(\mathcal{J}(g), \mathcal{J}(h))_{L^2(\Omega)} = (g, h)_{\mathbb{H}(\Gamma)}$. If the covariance function Γ can be expressed as

$$\Gamma(t_1, t_2) = \int_U f_{t_1}(u) f_{t_2}(u) d\nu(u), \quad (6.3.66)$$

where $(U, \mathcal{U}, \nu)$ is a measure space and $\{f_t : t \in T\} \subset L^2(U, \nu)$, then the RKHS $\mathbb{H}(\Gamma)$ is characterized by

$$\mathbb{H}(\Gamma) = \left\{ g : g(t) = \int_U g^*(u) f_t(u) d\nu(u), \text{ for some } g^* \in \overline{\text{span}}\{f_t, t \in T\} \right\}, \quad (6.3.67)$$

$$(g, h)_{\mathbb{H}(\Gamma)} = \int_U g^*(u) h^*(u) d\nu(u), \quad (6.3.68)$$

where $\overline{\text{span}}\{f_t, t \in T\}$ is the closure in $L^2(U, \nu)$ of all linear combinations of f_t , $t \in T$ (see Grenander [429]).

Let $\mathbb{H}_a(\Gamma^\kappa)$ be the RKHS of FBM B^κ on an interval $[0, a]$, where Γ^κ stands for the covariance function of B^κ . We shall now characterize the RKHS $\mathbb{H}_a(\Gamma^\kappa)$ of FBM.

Proposition 6.3.23 *Let $\kappa \in (-1/2, 1/2)$ and B^κ be a standard FBM. Then, the RKHS $\mathbb{H}_a(\Gamma^\kappa)$ of B^κ is*

$$\mathbb{H}_a(\Gamma^\kappa) = \left\{ g : g(t) = \sigma_\kappa^2 \int_0^a g^*(s) s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0,t)}(u))(s) ds, \text{ for some } g^* \in L^2[0, a] \right\} \quad (6.3.69)$$

with the inner product

$$(g, h)_{\mathbb{H}_a(\Gamma^\kappa)} = \sigma_\kappa^2 \int_0^a g^*(s) h^*(s) ds, \quad (6.3.70)$$

where σ_κ^2 is given in (6.2.20).

Proof By Proposition 6.2.8, we have

$$\Gamma^\kappa(t_1, t_2) = \sigma_\kappa^2 \int_0^a \left(s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, t_1)}(u))(s) \right) \left(s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, t_2)}(u))(s) \right) ds, \quad t_1, t_2 \in [0, a]. \quad (6.3.71)$$

Then, (6.3.69)–(6.3.70) would follow from (6.3.67)–(6.3.68). We just need to show that $\overline{\text{span}}\{f_t, t \in [0, a]\} = L^2[0, a]$, where $f_t(s) = s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, t)}(u))(s)$. The closure of the span is taken in $L^2[0, a]$ and hence $\overline{\text{span}}\{f_t, t \in [0, a]\} \subset L^2[0, a]$. The converse can be shown as follows. By Theorems 6.3.5 and 6.3.9, the set of elementary functions $\mathcal{E}$ is dense in Λ_a^κ . Elementary functions can be written as linear combinations of functions $1_{[0, t)}$, $t \in [0, a]$. Thus, for any function $g \in \Lambda_a^\kappa$, the function $f_g(s) = s^{-\kappa} (I_{a-}^\kappa u^\kappa g(u))(s)$ can be approximated in $L^2[0, a]$ by linear combinations of functions $f_t(s)$, $t \in [0, a]$. For any $b > 0$, there is a function $g = g_b$ such that $f_g = 1_{[0, b)}$: when $\kappa \in (0, 1/2)$, this follows from (6.3.14), and the case $\kappa \in (-1/2, 0)$ can be dealt with in a similar way. This shows that $\mathcal{E} \subset \overline{\text{span}}\{f_t, t \in [0, a]\}$. Since the linear span of indicator functions of intervals is dense in $L^2[0, a]$, it follows that $L^2[0, a] \subset \overline{\text{span}}\{f_t, t \in [0, a]\}$. $\square$

Example 6.3.24 (*Functions associated with increments of FBM*) By (6.3.65), the random variable $B^\kappa(t_2) - B^\kappa(t_1)$, $0 \leq t_1 < t_2 \leq a$, in $\mathcal{H}_a^\kappa$ is represented by the function $g(t) = \Gamma^\kappa(t, t_1) - \Gamma^\kappa(t, t_2)$ in $\mathbb{H}_a(\Gamma^\kappa)$. By (6.3.71), the corresponding function g^* is given by $g^*(s) = s^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[t_1, t_2)}(u))(s)$. Since the function g depends on κ , the RKHS can be viewed as representing “the space of integrals,” in contrast to the spaces introduced in Section 6.3.2, which can be regarded as “spaces of integrands” because they associate to $B^\kappa(t_2) - B^\kappa(t_1)$ the indicator function $1_{[t_1, t_2)}$ which plays the role of an integrand.

6.4 Applications

We provide here several applications illustrating the use of fractional Wiener integrals in addressing questions about FBM. These include Girsanov’s formula, the prediction formula and elementary linear filtering.

6.4.1 Girsanov’s Formula for FBM

Let $\{B^\kappa(u)\}_{u \in [0, a]}$ be a standard FBM and suppose that $\{M^\kappa(u)\}_{u \in [0, a]}$ is the fundamental martingale defined in Section 6.3.4. Suppose also that FBM is defined on some probability space $(\Omega, \mathcal{F}_a, P)$, where $\mathcal{F}_a$ is the σ -field generated by FBM B^κ on an interval $[0, a]$. Let

$$\mathcal{L}_a(M^\kappa) = \exp \left\{ M^\kappa(a) - \frac{1}{2} \langle M^\kappa, M^\kappa \rangle_a \right\} \quad (6.4.1)$$

and, for any $r \in \mathbb{R}$, define a probability measure P_r on $(\Omega, \mathcal{F}_a)$ by the relation

$$\left. \frac{dP_r}{dP} \right|_{\mathcal{F}_a} = \mathcal{L}_a(rM^\kappa) = \exp \left\{ rM^\kappa(a) - \frac{r^2}{2} \langle M^\kappa, M^\kappa \rangle_a \right\}. \quad (6.4.2)$$

Since $M^\kappa(a)$ is Gaussian with mean 0 and variance $\langle M^\kappa, M^\kappa \rangle_a$, we have $\mathbb{E} \mathcal{L}_a(rM^\kappa) = 1$ and therefore $\mathcal{L}_a(rM^\kappa)$ is well-defined. The following lemma is the key to Girsanov’s formula.

Lemma 6.4.1 Let $\{B^\kappa(u)\}_{u \in [0, a]}$ be a standard FBM with parameter $\kappa \in (-1/2, 1/2)$ and $\{M^\kappa(u)\}_{u \in [0, a]}$ be the fundamental martingale defined in Section 6.3.4. Then, for any $0 \leq s \leq t \leq a$,

$$\mathbb{E} M^\kappa(t) B^\kappa(s) = s. \quad (6.4.3)$$

Proof Suppose first that $\kappa \in (0, 1/2)$. Then, by (6.3.49), Proposition 6.3.7, the changes of variables $u = sy$ and $z = syx$ below, and the expression (6.2.20) for the constant σ_κ^2 , we obtain

$$\begin{aligned} (2\kappa + 1)\Gamma(1 + \kappa)\mathbb{E} M^\kappa(t) B^\kappa(s) &= \mathbb{E} \left(\int_0^a u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0, t)})(u) dB^\kappa(u) \int_0^a 1_{[0, s)}(u) dB^\kappa(u) \right) \\ &= \sigma_\kappa^2 \int_0^a z^{-\kappa} (I_{a-}^\kappa u^\kappa u^{-\kappa} (I_{a-}^{-\kappa} 1_{[0, t)})(u)) (z) z^{-\kappa} (I_{a-}^\kappa u^\kappa 1_{[0, s)}(u)) (z) dz \\ &= \sigma_\kappa^2 \int_0^a z^{-2\kappa} 1_{[0, t)}(z) (I_{a-}^\kappa u^\kappa 1_{[0, s)}(u)) (z) dz = \frac{\sigma_\kappa^2}{\Gamma(\kappa)} \int_0^s z^{-2\kappa} \int_z^s u^\kappa (u - z)^{\kappa-1} du dz \\ &= \frac{\sigma_\kappa^2}{\Gamma(\kappa)} \int_0^s du u^\kappa \int_0^u dz z^{-2\kappa} (u - z)^{\kappa-1} = \frac{\sigma_\kappa^2}{\Gamma(\kappa)} \int_0^1 dy \int_0^1 x^{-2\kappa} (1 - x)^{\kappa-1} dx \\ &= \frac{\sigma_\kappa^2}{\Gamma(\kappa)} B(\kappa, 1 - 2\kappa)s = (2\kappa + 1)\Gamma(1 + \kappa)s. \end{aligned}$$

When $\kappa \in (-1/2, 0)$, by using the integration by parts below, we have similarly,

$$\begin{aligned} (2\kappa + 1)\Gamma(1 + \kappa)\mathbb{E} M^\kappa(t) B^\kappa(s) &= \sigma_\kappa^2 \int_0^a z^{-2\kappa} 1_{[0, t)}(z) (I_{a-}^\kappa u^\kappa 1_{[0, s)}(u)) (z) dz \\ &= -\frac{\sigma_\kappa^2}{\Gamma(1 + \kappa)} \int_0^s z^{-2\kappa} \frac{d}{dz} \left(\int_z^s u^\kappa (u - z)^\kappa du \right) dz \\ &= \frac{(-2\kappa)\sigma_\kappa^2}{\Gamma(1 + \kappa)} \int_0^s z^{-2\kappa-1} \left(\int_z^s u^\kappa (u - z)^\kappa du \right) dz \\ &= \frac{(-2\kappa)\sigma_\kappa^2}{\Gamma(1 + \kappa)} \int_0^s du u^\kappa \int_0^u dz z^{-2\kappa-1} (u - z)^\kappa = \frac{(-2\kappa)\sigma_\kappa^2}{\Gamma(1 + \kappa)} s \int_0^1 dy \int_0^1 x^{-2\kappa-1} (1 - x)^\kappa dx \\ &= \frac{(-2\kappa)\sigma_\kappa^2}{\Gamma(1 + \kappa)} s B(-2\kappa, \kappa + 1) = (2\kappa + 1)\Gamma(1 + \kappa)s. \quad \square \end{aligned}$$

The following result is *Girsanov's formula* for FBM on an interval $[0, a]$.

Theorem 6.4.2 For $\kappa \in (-1/2, 1/2)$, let $\{B^\kappa(u)\}_{u \in [0, a]}$ be a standard FBM and, for $r \in \mathbb{R}$, let P_r be the probability measure defined by (6.4.2). Then the distribution of B^κ under P_r is equal to the distribution of fractional Brownian motion with a drift r , that is $\{B^\kappa(u) + ru\}_{u \in [0, a]}$, under P .

Proof It is enough to show that, for any $u_k \in [0, a]$ and $c_k \in \mathbb{C}$, $k = 1, \dots, n$,

$$\mathbb{E} \exp \left\{ \sum_{k=1}^n c_k B^\kappa(u_k) + r M^\kappa(a) - \frac{r^2}{2} \langle M^\kappa, M^\kappa \rangle_a \right\} = \mathbb{E} \exp \left\{ \sum_{k=1}^n c_k (B^\kappa(u_k) + ru_k) \right\}. \quad (6.4.4)$$

Observe that $\sum_{k=1}^n c_k B^\kappa(u_k) + r M^\kappa(a)$ is a complex-valued Gaussian random variable with zero mean and second moment $\mathbb{E}(\sum_{k=1}^n c_k B^\kappa(u_k) + r M^\kappa(a))^2$ equal to

$$\sum_{k,l=1}^n c_k c_l \Gamma^\kappa(u_k, u_l) + 2r \sum_{k=1}^n c_k u_k + r^2 \langle M^\kappa, M^\kappa \rangle_a,$$

where we used Lemma 6.4.1 and where Γ^κ denotes the covariance function of FBM B^κ . Then, since for a zero mean, complex-valued Gaussian random variable X , $\mathbb{E}e^{zX} = e^{(z^2 \mathbb{E}X^2)/2}$, $z \in \mathbb{C}$ (Exercise 6.10), the left-hand side of (6.4.4) becomes

$$\exp \left\{ \frac{1}{2} \sum_{k,l=1}^n c_k c_l \Gamma^\kappa(u_k, u_l) + r \sum_{k=1}^n c_k u_k \right\},$$

which is also the right-hand side of (6.4.4). $\square$

Remark 6.4.3 Theorem 6.4.2 naturally leads to the maximum likelihood estimator of the drift parameter r under the model $\{B^\kappa(u) + ru\}_{u \in [0,a]}$, namely,

$$\hat{r}_{mle} = \operatorname{argmax}_r \mathcal{L}_a(r M^\kappa), \quad (6.4.5)$$

where the likelihood $\mathcal{L}_a$ is given in (6.4.1). Solving $\frac{\partial}{\partial r} \mathcal{L}_a(r M^\kappa) = 0$ yields

$$\hat{r}_{mle} = \frac{M^\kappa(a)}{\langle M^\kappa, M^\kappa \rangle_a}. \quad (6.4.6)$$

See Exercise 6.11 for some properties of the estimator (6.4.6).

6.4.2 The Prediction Formula for FBM

The prediction problem for FBM B^κ is to find an explicit expression, called a prediction formula, for the conditional expectation $X = \mathbb{E}(B^\kappa(t) | B^\kappa(s), s \in [0, a])$ with some fixed $0 < a < t$. As already indicated in Section 6.3.1, we have $X \in \mathcal{H}_a^\kappa = \overline{\operatorname{span}}\{B^\kappa(u), u \in [0, a]\}$ and therefore we expect that $X = \int_0^a f(u) dB^\kappa(u)$ for some function f .

The prediction formula can heuristically be derived as follows. For $s \in [0, t]$, let

$$B^0(s) = \int_0^t u^{-\kappa} (I_{t-}^{-\kappa} z^\kappa 1_{[0,s]}(z))(u) dB^\kappa(u) \quad (6.4.7)$$

and recall from Section 6.3.5 that B^0 is the usual Brownian motion. One expects that the relation (6.4.7) is invertible, namely,

$$B^\kappa(s) = \int_0^t u^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[0,s]}(z))(u) dB^0(u), \quad (6.4.8)$$

and that $\sigma\{B^\kappa(s), s \in [0, a]\} = \sigma\{B^0(s), s \in [0, a]\}$ up to sets of measure zero, where $\sigma\{B^\kappa(s), s \in [0, a]\}$ is the σ -field generated by the random variables $B^\kappa(s)$, $s \in [0, a]$. Then, since B^0 is the usual Brownian motion,

$$\begin{aligned} \mathbb{E}(B^\kappa(t) | B^\kappa(s), s \in [0, a]) &= \mathbb{E}\left(\int_0^t u^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[0,t]}(z))(u) dB^0(u) \middle| B^0(s), s \in [0, a]\right) \\ &= \int_0^a u^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[0,t]}(z))(u) dB^0(u). \end{aligned}$$

By setting $1_{[0,t)} = 1_{[0,a)} + 1_{[a,t)}$ in this last relation and then using (6.4.8), we get

$$\mathbb{E}(B^\kappa(t)|B^\kappa(s), s \in [0, a]) = B^\kappa(a) + \int_0^a u^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z))(u) dB^0(u).$$

Since

$$u^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z))(u) = u^{-\kappa} I_{a-}^\kappa \left(v^\kappa v^{-\kappa} I_{a-}^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z))(v) \right)(u)$$

and since from (6.3.20) and (6.3.25) for an integrable function f ,

$$\int_0^a f(u) dB^\kappa(u) = \int_0^a u^{-\kappa} (I_{a-}^\kappa z^\kappa f(z))(u) dB^0(u),$$

we expect that

$$\mathbb{E}(B^\kappa(t)|B^\kappa(s), s \in [0, a]) = B^\kappa(a) + \int_0^a u^{-\kappa} \left(I_{a-}^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z)) \right)(u) dB^\kappa(u).$$

These computations suggest the following prediction formula for FBM.

Theorem 6.4.4 Let $0 < a < t$, $\kappa \in (-1/2, 1/2)$ and B^κ be FBM. Then

$$\mathbb{E}(B^\kappa(t)|B^\kappa(s), s \in [0, a]) = B^\kappa(a) + \int_0^a \Psi_t(a, u) dB^\kappa(u), \quad (6.4.9)$$

where, for $u \in (0, a)$,

$$\Psi_t(a, u) = u^{-\kappa} \left(I_{a-}^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z)) \right)(u) \quad (6.4.10)$$

$$= \frac{\sin(\kappa\pi)}{\pi} u^{-\kappa} (a-u)^{-\kappa} \int_a^t \frac{z^\kappa (z-a)^\kappa}{z-u} dz. \quad (6.4.11)$$

Remark 6.4.5 After the change of variables $z = t - (t-a)y$, the kernel function $\Psi_t(a, u)$ can be expressed as

$$\begin{aligned} \Psi_t(a, u) &= \frac{\sin(\kappa\pi)}{\pi} u^{-\kappa} (a-u)^{-\kappa} \int_0^1 \frac{(t - (t-a)y)^\kappa (t-a - (t-a)y)^\kappa}{t-u - (t-a)y} (t-a) dy \\ &= \frac{\sin(\kappa\pi)}{\pi} u^{-\kappa} (a-u)^{-\kappa} (t-a)^{\kappa+1} t^\kappa (t-u)^{-1} \\ &\quad \times \int_0^1 \left(1 - \frac{(t-a)}{t} y \right)^\kappa (1-y)^\kappa \left(1 - \frac{(t-a)}{t-u} y \right)^{-1} dy. \end{aligned}$$

By using Formula 3.211 in Gradshteyn and Ryzhik [421], p. 318, and using the fact that $B(\kappa+1, 1) = \Gamma(\kappa+1)\Gamma(1)/\Gamma(\kappa+2) = 1/(\kappa+1)$ (appearing in the formula), we get

$$\begin{aligned} \Psi_t(a, u) &= \frac{\sin(\kappa\pi)}{\pi(\kappa+1)} u^{-\kappa} (a-u)^{-\kappa} (t-a)^{\kappa+1} t^\kappa (t-u)^{-1} F_1 \left(1, -\kappa, 1, \kappa+2; \frac{(t-a)}{t}, \frac{(t-a)}{t-u} \right), \end{aligned} \quad (6.4.12)$$

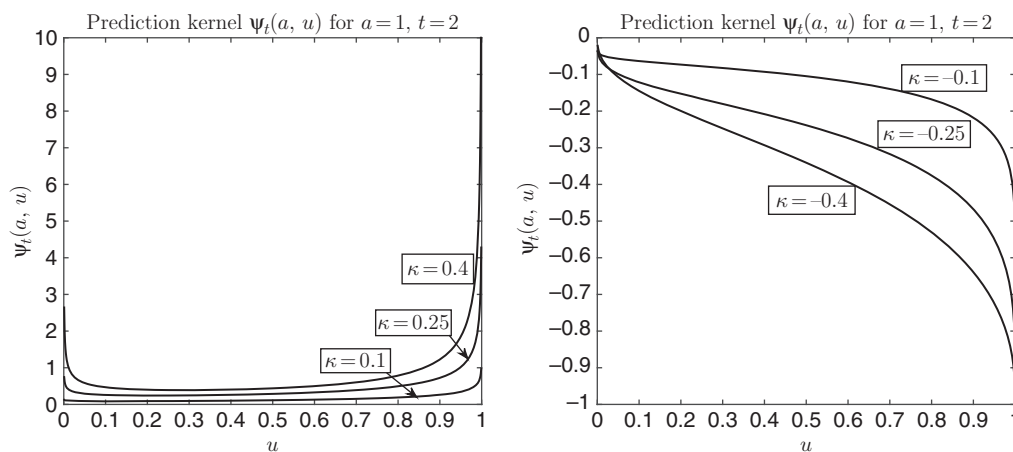


Figure 6.3 The kernel function $\Psi_t(a, u)$ in (6.4.12) for several values of κ for $a = 1$ and $t = 2$.

where F_1 is the so-called Appell's hypergeometric function. Figure 6.3 depicts the function $\Psi_t(a, u)$ computed through the formula (6.4.12).⁶

Remark 6.4.6 If $\kappa = 0$, namely if B^κ is the Brownian motion B^0 , then (6.4.10) becomes $\Psi_t(a, u) = 1_{[a,t)}(u)$ and hence $\int_0^a \Psi_t(a, u) dB^0(u) = 0$. Thus, as expected,

$$\mathbb{E}(B^0(t)|B^0(s), s \in [0, a]) = B^0(a).$$

Proof Let us first verify that the right-hand sides of (6.4.10) and (6.4.11) are equal when $\kappa \in (0, 1/2)$. Observe that by (6.1.3), for $v \in (0, a)$,

$$(I_{t-}^\kappa z^\kappa 1_{[a,t)}(z))(v) = \frac{1}{\Gamma(\kappa)} \int_0^t z^\kappa 1_{[a,t)}(z) (z-v)^{\kappa-1} dz = \frac{1}{\Gamma(\kappa)} \int_a^t z^\kappa (z-v)^{\kappa-1} dz \quad (6.4.13)$$

and hence by (6.1.12),

$$\begin{aligned} \left(I_{a-}^{-\kappa} (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z)) \right) (u) &= -\frac{1}{\Gamma(1-\kappa)\Gamma(\kappa)} \frac{d}{du} \int_u^a \int_a^t z^\kappa (z-v)^{\kappa-1} dz (v-u)^{-\kappa} dv \\ &= -\frac{\sin(\kappa\pi)}{\pi} \frac{d}{du} \int_a^t dz z^\kappa \int_u^a dv (z-v)^{\kappa-1} (v-u)^{-\kappa}, \end{aligned} \quad (6.4.14)$$

since $\Gamma(1-\kappa)\Gamma(\kappa) = \pi/\sin(\kappa\pi)$. By making the change of variables $v = u + (z-u)s$ and then taking d/du inside the integral, we get

⁶ The Appell hypergeometric function is implemented in the Matlab package “Generation of Random Variates,” or in the R package “appell.”

$$\begin{aligned}
\left(I_{a-}^{-\kappa}(I_{t-}^{\kappa}z^{\kappa}1_{[a,t)}(z))\right)(u) &= -\frac{\sin(\kappa\pi)}{\pi}\frac{d}{du}\int_a^t dz z^{\kappa}\int_0^{\frac{a-u}{z-u}} ds (1-s)^{\kappa-1}s^{-\kappa} \\
&= -\frac{\sin(\kappa\pi)}{\pi}\int_a^t dz z^{\kappa}\left(1-\frac{a-u}{z-u}\right)^{\kappa-1}\left(\frac{a-u}{z-u}\right)^{-\kappa}\frac{d}{du}\left(\frac{a-u}{z-u}\right) \\
&= \frac{\sin(\kappa\pi)}{\pi}(a-u)^{-\kappa}\int_a^t \frac{z^{\kappa}(z-a)^{\kappa}}{z-u}dz. \tag{6.4.15}
\end{aligned}$$

When $\kappa \in (-1/2, 0)$, the right-hand sides of (6.4.10) and (6.4.11) are still equal. To see this, start with (6.1.12) or use Remark 6.1.6. In the latter case, observe first that (6.4.13) with $\Gamma(\kappa) = \Gamma(1+\kappa)/\kappa$ still holds for $v \in (0, a)$, since $z \neq v$ in the range of integration. Then, (6.4.14) becomes

$$\begin{aligned}
\left(I_{a-}^{-\kappa}(I_{t-}^{\kappa}z^{\kappa}1_{[a,t)}(z))\right)(u) &= \frac{\kappa}{\Gamma(1+\kappa)\Gamma(-\kappa)}\int_u^a \int_a^t z^{\kappa}(z-v)^{\kappa-1}dz (v-u)^{-\kappa-1}dv \\
&= -\frac{\kappa \sin(\kappa\pi)}{\pi}\int_a^t dz z^{\kappa}\int_u^a dv (z-v)^{\kappa-1}(v-u)^{-\kappa-1}.
\end{aligned}$$

Observe now that, by making the change of variables $v = u + (z-u)s$ and then $s = 1/w$,

$$\begin{aligned}
\kappa \int_u^a (z-v)^{\kappa-1}(v-u)^{-\kappa-1}dv &= \kappa(z-u)^{-1}\int_0^{\frac{a-u}{z-u}} (1-s)^{\kappa-1}s^{-\kappa-1}ds \\
&= \kappa(z-u)^{-1}\int_{\frac{z-u}{a-u}}^{\infty} (w-1)^{\kappa-1}dw = -(z-u)^{-1}\left(\frac{z-u}{a-u}-1\right)^{\kappa} = -\frac{(a-u)^{-\kappa}(z-a)^{\kappa}}{z-u}.
\end{aligned}$$

This yields the relation (6.4.15) in the case $\kappa \in (-1/2, 0)$ as well.

To verify that the integral in (6.4.9) is well-defined for $\kappa \in (-1/2, 0)$, it is enough to check that the function $s^{-\kappa}(I_{t-}^{\kappa}z^{\kappa}1_{[a,t)}(z))(s)$ belongs to $L^2[0, a]$. The same condition needs to be verified in the case $\kappa \in (0, 1/2)$. In both cases this can be deduced from (6.4.13). Finally, one needs to prove (6.4.9), namely that the best predictor of $B^{\kappa}(t)$, $t > a$, given $\{B^{\kappa}(s), s \in [0, a]\}$ is $\widehat{\xi}_t = B^{\kappa}(a) + \int_0^a \Psi_t(a, u)dB^{\kappa}(u)$; that is, $\mathbb{E}(B^{\kappa}(t)|B^{\kappa}(s), s \in [0, a]) = \widehat{\xi}_t$. By the definition of conditional expectation, this is equivalent to $\mathbb{E}B^{\kappa}(t)1_A = \mathbb{E}\widehat{\xi}_t1_A$ for any $A \in \sigma\{B^{\kappa}(s), s \in [0, a]\}$. Since the σ -field $\sigma\{B^{\kappa}(s), s \in [0, a]\}$ is generated by cylinder sets $C = \{B^{\kappa}(s_j) \in B_j, j = 1, \dots, n, s_j \in [0, a], B_j \in \mathcal{B}(\mathbb{R})\}$, it is enough to show that $\mathbb{E}B^{\kappa}(t)1_C = \mathbb{E}\widehat{\xi}_t1_C$. Since the vectors $(B^{\kappa}(t), B^{\kappa}(s_j), j = 1, \dots, n)$ and $(\widehat{\xi}_t, B^{\kappa}(s_j), j = 1, \dots, n)$ are zero mean Gaussian, the last relation follows from $\mathbb{E}B^{\kappa}(t)B^{\kappa}(s) = \mathbb{E}\widehat{\xi}_tB^{\kappa}(s)$, for all $s \in [0, a]$, or equivalently,

$$\mathbb{E}B^{\kappa}(s)(B^{\kappa}(t) - B^{\kappa}(a)) = \mathbb{E}B^{\kappa}(s)\int_0^a \Psi_t(a, u)dB^{\kappa}(u)$$

or

$$\mathbb{E}\int_0^t 1_{[0,s)}(u)dB^{\kappa}(u)\int_0^t 1_{[a,t)}(u)dB^{\kappa}(u) = \mathbb{E}\int_0^a 1_{[0,s)}(u)dB^{\kappa}(u)\int_0^a \Psi_t(a, u)dB^{\kappa}(u). \tag{6.4.16}$$

By Definitions 6.3.6 and 6.3.10, the left-hand side of (6.4.16) is

$$\sigma_{\kappa}^2 \int_0^t v^{-2\kappa}(I_{t-}^{\kappa}u^{\kappa}1_{[0,s)}(u))(v)(I_{t-}^{\kappa}u^{\kappa}1_{[a,t)}(u))(v)dv$$

or

$$\sigma_\kappa^2 \int_0^a v^{-2\kappa} (I_{a-}^\kappa u^\kappa 1_{[0,s)}(u))(v) (I_{t-}^\kappa u^\kappa 1_{[a,t)}(u))(v) dv,$$

since $(I_{t-}^\kappa u^\kappa 1_{[0,s)}(u))(v) = 0$ for $a < v < t$ and $(I_{t-}^\kappa u^\kappa 1_{[0,s)}(u))(v) = (I_{a-}^\kappa u^\kappa 1_{[0,s)}(u))(v)$ for $v \in [0, a]$ by using Definitions 6.1.1 and 6.1.4. Similarly, the right-hand side of (6.4.16) is

$$\sigma_\kappa^2 \int_0^a v^{-2\kappa} (I_{a-}^\kappa u^\kappa 1_{[0,s)}(u))(v) (I_{a-}^\kappa u^\kappa \Psi_t(a, u))(v) dv,$$

which equals the left-hand side, since $(I_{a-}^\kappa u^\kappa \Psi_t(a, u))(v) = (I_{t-}^\kappa z^\kappa 1_{[a,t)}(z))(v)$ by using the expression (6.4.10). $\square$

6.4.3 Elementary Linear Filtering Involving FBM

Fractional integrals and derivatives can be also used to solve a filtering problem in a simple linear system driven by FBM. To describe the model, we let $\{B^\kappa(u)\}_{u \in [0,t]}$, $t > 0$, $\kappa \in (-1/2, 1/2)$, be a standard FBM and ξ be a fixed random variable, called a signal, which is independent of B^κ and is normally distributed with mean 0 and variance 1. Suppose that the observation process $\{Y(v)\}_{v \in [0,t]}$ is given by

$$Y(v) = \int_0^v a(u) \xi du + \int_0^v b(u) dB^\kappa(u), \quad (6.4.17)$$

where a and b are some known (deterministic) functions and where B^κ is a source of noise. The filtering problem is then to compute the best estimator $\widehat{\xi}_t$ of the signal ξ based on the observations $Y(v)$, $v \in (0, t)$. “Best” here means that

$$\mathbb{E}(\widehat{\xi}_t - \xi)^2 \leq \mathbb{E}(\widetilde{\xi}_t - \xi)^2 \quad (6.4.18)$$

for any other estimator $\widetilde{\xi}_t$ of ξ based on the same observations. It is well-known that (6.4.18) implies that $\widehat{\xi}_t = \mathbb{E}(\xi | Y(v), v \in [0, t])$ and, since the process $(\xi, \{Y(v)\}_{v \in [0,t]})$ is Gaussian, we expect that

$$\widehat{\xi}_t = \mathbb{E}(\xi | Y(v), v \in [0, t]) = \int_0^t \widehat{f}_t(u) dY(u) \quad (6.4.19)$$

for some deterministic function $\widehat{f}_t$. The problem is then to find the function $\widehat{f}_t$.

We will find the function $\widehat{f}_t$ in (6.4.19) heuristically, in the sense that we will assume that all integrals involved are well-defined. As in Section 6.4.2, (6.4.19) is equivalent to $\mathbb{E}(\int_0^t \widehat{f}_t(u) dY(u) - \xi | Y(v), v \in [0, t]) = 0$ and this holds if and only if

$$\mathbb{E}\left(\int_0^t \widehat{f}_t(u) dY(u) - \xi\right)(Y(t_2) - Y(t_1)) = 0, \quad (6.4.20)$$

for all $0 \leq t_1 < t_2 \leq t$. Since

$$\begin{aligned} \int_0^t \widehat{f}_t(u) dY(u) - \xi &= \left(\int_0^t \widehat{f}_t(u) a(u) du - 1\right) \xi + \int_0^t \widehat{f}_t(u) b(u) dB^\kappa(u), \\ Y(t_2) - Y(t_1) &= \int_0^t a(u) 1_{[t_1, t_2)}(u) du \xi + \int_0^t b(u) 1_{[t_1, t_2)}(u) dB^\kappa(u), \end{aligned}$$

we obtain from the independence of ξ and B^κ that (6.4.20) is equivalent to

$$\begin{aligned} & \left(\int_0^t \widehat{f}_t(u) a(u) du - 1 \right) \int_0^t a(u) 1_{[t_1, t_2]}(u) du \\ & + \mathbb{E} \int_0^t \widehat{f}_t(u) b(u) d B^\kappa(u) \int_0^t b(u) 1_{[t_1, t_2]}(u) d B^\kappa(u) = 0 \end{aligned} \quad (6.4.21)$$

or to

$$\begin{aligned} & \sigma_\kappa^2 \int_0^t s^{-2\kappa} \left(I_{t-}^\kappa u^\kappa \widehat{f}_t(u) b(u) \right)(s) \left(I_{t-}^\kappa u^\kappa b(u) 1_{[t_1, t_2]}(u) \right)(s) ds \\ & = \left(1 - \int_0^t \widehat{f}_t(u) a(u) du \right) \int_0^t a(u) 1_{[t_1, t_2]}(u) du. \end{aligned} \quad (6.4.22)$$

By using the fractional integration by parts formula (6.1.6), we can write (6.4.22) as

$$\begin{aligned} & \sigma_\kappa^2 \int_0^t \left(I_{0+}^\kappa s^{-2\kappa} (I_{t-}^\kappa u^\kappa \widehat{f}_t(u) b(u))(s) \right)(z) z^\kappa b(z) 1_{[t_1, t_2]}(z) dz \\ & = \int_0^t a(z) \left(1 - \int_0^t \widehat{f}_t(u) a(u) du \right) 1_{[t_1, t_2]}(z) dz. \end{aligned} \quad (6.4.23)$$

We expect (6.4.23) to hold for all $0 \leq t_1 < t_2 \leq t$ if and only if

$$\begin{aligned} & \sigma_\kappa^2 z^\kappa b(z) \left(I_{0+}^\kappa s^{-2\kappa} (I_{t-}^\kappa u^\kappa \widehat{f}_t(u) b(u))(s) \right)(z) \\ & = a(z) \left(1 - \int_0^t \widehat{f}_t(u) a(u) du \right). \end{aligned} \quad (6.4.24)$$

To solve (6.4.24), one first finds $\widehat{f}_t^*$ such that

$$\sigma_\kappa^2 z^\kappa b(z) \left(I_{0+}^\kappa s^{-2\kappa} (I_{t-}^\kappa u^\kappa \widehat{f}_t^*(u) b(u))(s) \right)(z) = a(z). \quad (6.4.25)$$

By inverting (6.4.25), it is easily seen that

$$\widehat{f}_t^*(u) = \sigma_\kappa^{-2} u^{-\kappa} b(u)^{-1} \left(I_{t-}^{-\kappa} s^{2\kappa} (I_{0+}^{-\kappa} z^{-\kappa} a(z) b(z)^{-1})(s) \right)(u). \quad (6.4.26)$$

The function $\widehat{f}_t$ satisfying (6.4.24) is then given by

$$\widehat{f}_t(u) = P(t) \widehat{f}_t^*(u), \quad (6.4.27)$$

where

$$P(t) = \left(1 + \int_0^t \widehat{f}_t^*(u) a(u) du \right)^{-1}. \quad (6.4.28)$$

Indeed, after substituting (6.4.27) into (6.4.24), one gets

$$P(t) = 1 - P(t) \int_0^t \widehat{f}_t^*(u) a(u) du,$$

which leads to (6.4.28). See Section 7.4 for references on filtering involving fractional Brownian motion.

6.5 Exercises

The symbols * and ** next to some exercises are explained in Section 2.12.

Exercise 6.1 Prove the relation (6.1.35) by direct computations using Definition 6.1.18.

Exercise 6.2 Prove the relation (6.1.41) by direct computations. *Hint:* Use (6.1.35).

Exercise 6.3 Prove the relation (6.2.25).

Exercise 6.4 In the proof of Proposition 6.3.2, show that the definition of $\mathcal{I}_a^\kappa(f)$ does not depend on an approximating sequence $\{f_n\}$ of f .

Exercise 6.5 Show that the relation (6.3.7) holds for Wiener integrals $\mathcal{I}_a^0(f)$ and $\mathcal{I}_a^0(g)$ defined with respect to the usual (standard) Brownian motion and $f, g \in \mathcal{E}$.

Exercise 6.6 (i) Show that the function (6.3.13) satisfies the relation (6.3.14). (ii) Show the statement leading to (6.3.16).

Exercise 6.7* Show the convergence of $I_{i,h}$, $i = 1, \dots, 5$, stated following (6.3.45) and under the assumption that f is γ -Hölder continuous with $0 < (-\kappa) < \gamma$.

Exercise 6.8 Consider the fractional OU process Y^κ in Example 6.3.22. (i) Show that it satisfies the equation (6.3.64). (ii) Show that it is long-range dependent with parameter $d = \kappa$ in the sense that $\mathbb{E}Y^\kappa(0)Y^\kappa(h) \sim Ch^{2\kappa-1}$, as $h \rightarrow \infty$. *Hint:* See Cheridito et al. [229].

Exercise 6.9** Various spaces of integrands were introduced on an interval $[0, a]$ in Section 6.3.2, and on the real line $\mathbb{R}$ in Section 6.3.6. How are these spaces related? For example, if f is a function on $[0, a]$ belonging to one of the spaces, does this imply that the function $f1_{[0,a]}$ defined on $\mathbb{R}$ belongs to one of the spaces for $\mathbb{R}$? And vice versa, if f is a function on $\mathbb{R}$ belonging to one of the spaces, what about its restriction $f|_{[0,a]}$ to the interval $[0, a]$?

Exercise 6.10 Suppose that $X = X_1 + iX_2$ is a zero mean, complex-valued Gaussian random variable (that is, the random bivariate vector $(X_1, X_2)'$ is Gaussian and has zero mean). Show that $\mathbb{E}e^{zX} = e^{(z^2\mathbb{E}X^2)/2}$, $z \in \mathbb{C}$.

Exercise 6.11 Consider the maximum likelihood estimator $\widehat{r}_{mle}$ in (6.4.6). Show that, under the probability measure P_r defined by (6.4.2), the estimator $\widehat{r}_{mle}$ has the following distribution:

$$\widehat{r}_{mle} \stackrel{d}{=} \mathcal{N}\left(r, \frac{1}{ca^{1-2\kappa}}\right)$$

for a constant c (depending on κ).

6.6 Bibliographical Notes

Section 6.1: A comprehensive reference on fractional integrals and derivatives is the monograph by Samko et al. [878] (consisting of 976 pages and with parts of it in tiny font!). For a geometric interpretation of fractional calculus, see Podlubny [826]. The presentation found in Section 6.1 is motivated by applications to fractional Brownian motions, and is aimed to be concise, self-contained and intuitive.

Section 6.2: The representation of FBM on the real line and the connection to fractional integrals (derivatives) in Section 6.2.2 is known already from the celebrated work of Mandelbrot and Van Ness [682], who were one of the first to consider and certainly to popularize FBM. The term “fractional” refers precisely to the fact that FBM can be obtained from Brownian motion through fractional integration (derivation).

The representation of FBM on an interval in Section 6.2.1 appeared in Molchan [726], Molchan and Golosov [725]. See also Molchan [727] with a discussion on earlier developments. It was later rediscovered by Barton and Poor [104], Norros et al. [758], Decreusefond and Üstünel [296].

Connections between the representations of FBM on the real line and an interval are studied in Jost [537]. See also Picard [809].

Section 6.3: This section is based on the work in Pipiras and Taqqu [815, 817, 818]. Fractional Wiener integrals are also considered in many of the references to Section 7.1 on stochastic integration given below. Integrands for FBM are viewed as generalized functions in Jolis (2007), forming a Hilbert space even in the case $H > 1/2$.

RKHS associated with general self-similar processes and several applications are considered in Nuzman and Poor [775]. Similar integration theory for the so-called tempered FBM is developed in Meerschaert and Sabzikar [708]. Other work on fractional OU process considered in Example 6.3.22 can be found in Kaarakka and Salminen [543], Sun and Guo [933].

Section 6.4: The Girsanov formula for FBM and its use in statistical estimation of the drift parameter (see Remark 6.4.3) appeared in Norros et al. [758]. Extensions of the formula to more general drift processes with applications to statistical inference can be found in Le Breton [600], Kleptsyna et al. [566, 566], Brouste and Kleptsyna [190], and the monograph by Prakasa Rao [831], among others. See also Sottinen [915].

The prediction formula for FBM appeared in Molchan [726], Molchan and Golosov [725], Gripenberg and Norros [433]. For extensions, see Duncan [334], Duncan and Fink [336], Fink et al. [364].

Linear filtering problems for FBM were considered in Le Breton [600], Kleptsyna and co-authors [565, 566, 564], Ahmed and Charalambous [16], Inoue et al. [522] and others.

Other problems which involve fractional Wiener integrals, Gaussian spaces and spaces of integrands concern local independence of FBM (Norros and Saksman [757]), continuity in the Hurst parameter of the law of fractional Wiener integrals (Jolis and Viles [536]), mutual information for stochastic signals and FBM (Duncan [335]), fractional Brownian bridges (Gasbarra et al. [383]).

Other notes: In this chapter, we explored connections between FBM and fractional calculus. Fractional calculus is also related in interesting and fundamental ways to stable distributions, stable Lévy motions and the so-called anomalous diffusions. A basic connection is that the transition probability $p_t(x)$ of a stable Lévy motion satisfies a space-time fractional diffusion equation. See, for example, Meerschaert [707], Meerschaert and Sikorskii [712].